SlotATR: Toward Object-Centric Learning for SAR Target Recognition

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***Abstract*—Synthetic aperture radar (SAR) images are inherently affected by complex and heterogeneous clutter, whose scattering characteristics often overlap with those of targets and vary across sensing conditions. This leads to severe feature entanglement and degraded generalization in automatic target recognition (ATR), particularly under varying clutter distributions and signal-to-clutter ratio (SCR). To address this challenge, we propose SlotATR, an object-centric learning (OCL)-based ATR framework that disentangles target representations from background clutter in SAR images without requiring mask annotations, and performs ATR using the isolated target features. SlotATR consists of two stages. In the first stage, SlotSAR introduces a grouping feature refinement module (GFRM), which integrates semantic and scattering features to refine the feature space referenced during slot updates. This design facilitates more reliable slot competition, leading to improved object-centric representations. In the second stage, SlotCLS utilizes the disentangled target slot representations to produce target-centric features for ATR classification under cluttered conditions. Extensive experiments on ATRNet-STAR under diverse clutter conditions demonstrate that the proposed framework consistently improves both object-centric decomposition and ATR performance, particularly under challenging clutter variations. These results indicate that SlotATR enhances target–clutter separability in SAR imagery, providing a practical approach toward object-centric SAR image analysis in remote sensing.**

***Index Terms*—Automatic target recognition (ATR), feature-level clutter reduction (FLCR), object-centric learning (OCL), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) imagery offers high-resolution observations under all-day, all-weather, and long-range conditions, and it has been widely applied in both military and civilian fields [1], [2]. As a fundamental and challenging task in remote sensing, SAR automatic target recognition (SAR ATR) aims to automatically detect and classify objects of interest, such as vehicles, ships, or aircraft, playing a vital role in surveillance, disaster assessment, and emergency response [3], [4], [5], [6], [7], [8]. Recent advances in SAR automatic target recognition (ATR) have been driven by the emergence of large-scale datasets [9], [10], [11] and foundation models [12], [13], [14].

Despite these advances, SAR ATR performance still degrades significantly when there is a distribution shift between training and test environments, or when unseen clutter conditions are encountered. SAR images contain both targets and nontarget components, such as background clutter from terrain, man-made structures, or vegetation, and speckle noise from sensor characteristics. In particular, due to the nonuniform scattering of ground surfaces, clutter can often resemble target signatures in intensity and pattern [15]. As a result, current models tend to focus on clutter regions and learn spurious features, leading to degraded generalization under unseen clutter conditions [9], [16], [17], [18], [19] [Fig. 1(a)]. Consequently, disentangling target features from clutter has become a key challenge for improving SAR ATR robustness.

To address this challenge, various approaches have been proposed. One line of research focuses on preprocessing-based target region extraction, which isolates target pixels prior to feature extraction [Fig. 1(b)]. Guo et al. [19], Choi et al. [20], and Zhou et al. [21] are sensitive to the intensity statistics of SAR imagery and often assume that the target is located near the image center, limiting their applicability in practical classification scenarios. Methods based on target scatterers, such as [22], [23], and [24], encode structured target information by extracting scatterer representations, but suffer from hyperparameter sensitivity and high computational cost. More fundamentally, the heavy reliance on hyperparameters makes these methods fragile under unseen clutter conditions: when the assumed intensity thresholds no longer hold, target information is lost before reaching the model, directly degrading recognition performance.

These limitations have led to the introduction of feature-level clutter reduction (FLCR), where the model learns to reduce clutter interference directly in the feature space without relying on preprocessing pipelines [Fig. 1(c)]. Methods such as [16], [17], [25] achieve more robust classification by learning clutter-invariant representations [18], [26] or reducing background responses at the feature level [16], [17], [25]. However, existing studies mostly rely on supervised learning with precise pixel-level target masks, which are rarely available in SAR imagery and costly to generate with consistent quality. This

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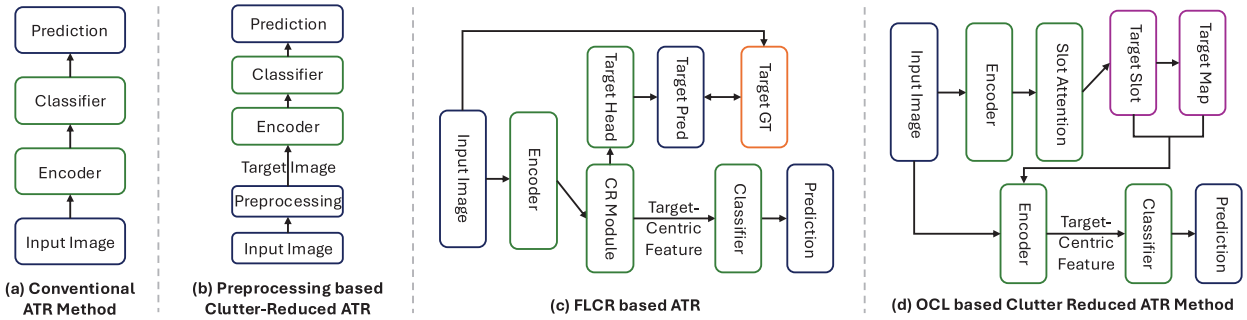


Fig. 1. Comparison of SAR ATR frameworks. (a) Conventional ATR encodes input images for classification [27], [28]. (b) Preprocessing-based methods extract target regions before encoding [20], [21]. (c) FLCR-based methods use pixel-level target GT masks to learn clutter-reduced features [16], [17]. (d) Proposed OCL-based method extracts target slots and attention maps without mask annotations, enabling clutter-reduced classification.

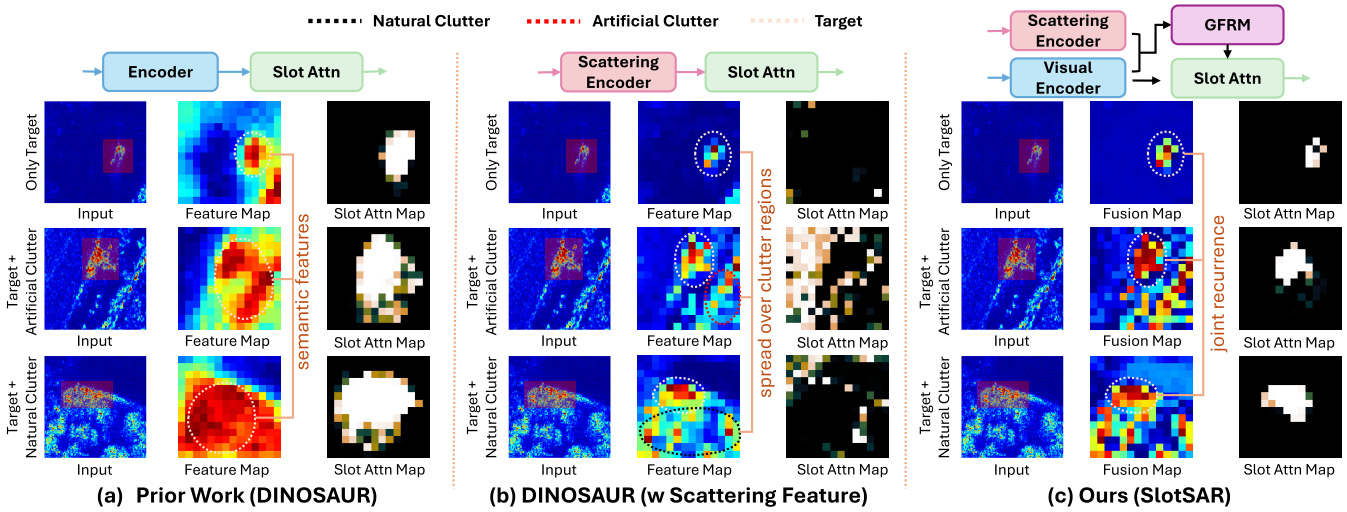


Fig. 2. Overview of SlotSAR. (a) Prior work shows overlapping target–clutter responses in both feature and slot attention maps. (b) Adding scattering features lacks sufficient semantic cues for stable OCL. (c) SlotSAR integrates semantic and scattering information via GFRM, enabling slot attention to group recurring visual and scattering patterns into object-centric representations. The red regions indicate the ground-truth target regions.

dependence on annotations limits their applicability in real-world scenarios.

To address these limitations, we propose SlotATR, an object-centric learning (OCL) based framework for target-centric SAR ATR without any mask annotations [Fig. 1(d)]. SlotATR consists of two stages: In the first stage, we apply OCL to disentangle target and clutter representations from raw SAR features without mask annotations (SlotSAR). In the second stage, the resulting target-centric slot representations are integrated into ATR backbone architectures to enable target-centric classification (SlotCLS).

A. OCL for SAR Imagery

OCL aims to decompose visual scenes into structured representations of individual objects without manual supervision [29]. A representative method is slot attention [30], which clusters recurring visual patterns into object-centric representations via cross-attention between input features and latent slots. In optical imagery, rich visual cues such as color, texture, and object continuity naturally induce intraobject feature homogeneity, providing stable competition signals that guide each slot toward a distinct object [31]. Building on this, DINOSAUR [31] introduces an inductive bias by reconstructing features with high intraobject homogeneity, enabling slot attention to scale to real-world optical scenes.

However, this inductive bias is less effective for SAR imagery. The absence of color information [32], [33] and unstable textures caused by speckle noise cause targets and clutter to exhibit similar high-level feature distributions [34], [35]. Consequently, feature homogeneity extends beyond individual object boundaries and becomes shared across target and clutter regions, resulting in limited interclass separability even in features extracted from an SAR foundation model. Such limited separability can weaken the competition among slots during object grouping. As illustrated in Fig. 2(a), relying solely on visual features makes it difficult for slots to converge to distinct target and clutter representations.

To address this limitation, we refine the feature space used for object grouping by jointly exploiting visual and scattering features. Wavelet scattering features characterize the local distribution of signal energy across multiple scales and orientations [36], [37], [38] and effectively capture high local energy responses induced by strong backscattering structures in SAR imagery [39], [40]. Thus, as illustrated in Fig. 2(b), wavelet scattering features can emphasize dominant scattering responses that are less pronounced in high-level visual features. While visual features provide high-level semantic infor-

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mation, scattering features additionally capture local scattering responses and their spatial distributions. Integrating these complementary features enriches the visual representation with local scattering responses and their spatial distributions that are weakly expressed by visual features alone. This allows slot attention to exploit both high-level semantic information and local scattering responses for grouping target and clutter regions. By jointly aggregating recurring visual and scattering patterns, slot attention can form object-centric representations in SAR imagery. Among various auxiliary features, wavelet scattering features most effectively improve interclass separability in the grouping feature space (Section IV-C1).

Recent SAR target recognition studies have explored various feature fusion strategies to exploit the complementary properties of visual and scattering features. For example, these methods fuse the two features at the encoder or decision level to enhance target representations, thereby improving classification performance [41], [42], [43], [44], [45], and domain alignment [46], [47], or facilitate the generation of physically consistent SAR imagery [48], [49]. However, exploiting such complementary features to disentangle targets from clutter through OCL remains underexplored. In this work, we investigate the integration of visual and scattering information within the slot attention process, rather than at the encoder or decision level, for OCL in SAR imagery.

Specifically, we introduce SlotSAR, a framework that disentangles target representations from background clutter in SAR imagery without mask annotations. Rather than modifying the slot attention mechanism, SlotSAR introduces a grouping feature refinement module (GFRM) that refines the feature space referenced during slot updates by exploiting semantic and scattering features. This design encourages explicit competition between target and clutter slots, providing sufficient signals for learning object-centric representations without mask supervision, as illustrated in Fig. 2(c).

B. Clutter Reduced SAR ATR

While SlotSAR focuses on separability between target and clutter regions, ATR requires a further level of discrimination: interclass separability among target categories within the isolated target region. However, conventional ATR backbones process feature maps without explicit awareness of target and clutter regions, making them susceptible to clutter interference under unseen conditions. A straightforward approach is to directly inject the target attention map into intermediate feature maps via element-wise multiplication [16], [17], reducing background responses at the feature level. However, this approach applies uniform reduction across all feature channels, failing to account for the channel-wise importance of target-related features.

To address this, we propose SlotCLS, which integrates the slot representations by SlotSAR into ATR backbone architectures through a dual attention mechanism. Specifically, SlotCLS employs the target attention map as spatial attention to localize target regions within the feature map, while the target slot vector is used as channel attention to emphasize target-relevant feature channels. By jointly applying spatial and channel attention derived from object-centric slot representations, SlotCLS produces target-centric feature representations that mitigate clutter interference, enabling more discriminative classification without requiring any mask annotations.

The main contributions of this article are as follows.

1) We analyze the inter and intraobject feature variance in SAR imagery and identify insufficient interclass separability as the primary cause of slot collapse onto clutter backgrounds.
2) We propose SlotSAR, which introduces a GFRM that refines the feature space referenced during slot updates by incorporating wavelet scattering features as auxiliary priors, enabling improved target–clutter disentanglement without mask annotations.
3) We design a dual-attention integration strategy, named the SlotCLS module, that integrates object-centric slot representations for improved target-centric classification.
4) We present SlotATR, a general two-stage framework that combines SlotSAR and SlotCLS, and validate its effectiveness under extended operating conditions (EOCs) with significant clutter variations.

II. Analysis of OCL in SAR Imagery

In this section, we analyze why obtaining robust object-centric representations is particularly challenging in SAR imagery. We begin with a brief overview of the slot attention mechanism, and then analyze feature maps and interclass variance across optical and SAR imagery.

A. Preliminaries

Slot attention [30] is a module that learns object-centric representations from perceptual input. Given an input image, an encoder first extracts spatial feature maps, which are then flattened into a set of L feature vectors $X_v \in \mathbb{R}^{L\times D_v}$. Slot attention then maps these features to a set of M slot vectors $S \in \mathbb{R}^{M\times D_{\text{slot}}}$, where each slot is intended to capture the properties of a distinct entity in the scene. This mapping is performed through an iterative attention mechanism in which slots compete to explain different parts of the input, encouraging each slot to specialize to a different region or object. The module consists of three stages: slot initialization, iterative attention computation, and slot update.

Slot Initialization: Given M slots forming $S \in \mathbb{R}^{M\times D_{\text{slot}}}$, the initial slots $S^{(0)}$ are sampled from a learnable Gaussian distribution via a reparameterization trick.

Attention Computation: The key, value, and query representations are obtained as

$$K_v = W^k \cdot \text{LayerNorm}(X_v), \quad V_v = W^v \cdot \text{LayerNorm}(X_v)$$
$$Q^{(r)} = W^q \cdot \text{LayerNorm}\left(S^{(r-1)}\right). \tag{1}$$

The attention logits $E^{(r)} \in \mathbb{R}^{L\times M}$ are computed as $E^{(r)} = K_v(Q^{(r)})^\top / \sqrt{d_h}$, and normalized across the slot axis to introduce competition between slots

$$\mathcal{A}^{(r)}_{i,j} = \frac{\exp\left(E^{(r)}_{i,j}\right)}{\sum_{l=1}^{M}\exp\left(E^{(r)}_{i,l}\right)} \tag{2}$$

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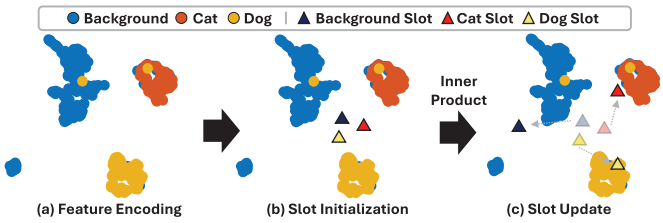


Fig. 3. Visualization of slot attention in feature space. (a) Input features are encoded, where each point represents a token-wise feature vector from a single input image, forming distinct clusters corresponding to background, cat, and dog regions. (b) Slots are randomly initialized in the feature space. (c) Each slot is updated via inner product-based attention, converging toward its corresponding object cluster.

where $i \in \{1, \ldots, L\}$ indexes input tokens and $j \in \{1, \ldots, M\}$ indexes slots.

Slot Update: The aggregated update for each slot is computed via a weighted mean over input features, where the attention weights are row-normalized: $\bar{\mathcal{A}}_{i,j}^{(r)} = (\mathcal{A}_{i,j}^{(r)}) / \left(\sum_{m=1}^{L} \mathcal{A}_{m,j}^{(r)}\right)$, and the slot states are updated via a gated recurrent unit (GRU) [50] followed by a residual MLP

$$\begin{aligned} \tilde{S}^{(r)} &= \mathrm{GRU}\left(S^{(r-1)},\ \left(\bar{\mathcal{A}}^{(r)}\right)^{\top} \cdot V_v\right) \\ S^{(r)} &= \tilde{S}^{(r)} + \mathrm{MLP}\left(\mathrm{LayerNorm}\left(\tilde{S}^{(r)}\right)\right). \end{aligned} \tag{3}$$

This process is repeated for R iterations, after which the slots converge to entity-centric representations of the input, where $\mathcal{A}$ encodes the spatial localization of each entity and S captures its semantic properties.

Early works built upon slot attention with CNN-based encoders, where the reconstruction objective groups together independently repeating visual patterns across the dataset [30]. However, such approaches rely heavily on low-level cues such as color to assign pixels to objects, and consequently struggle to scale beyond simple synthetic datasets. To address this limitation, DINOSAUR [31] introduced an additional inductive bias by replacing pixel-level reconstruction with self-supervised feature reconstruction, using features extracted from a pretrained DINO [51] model as the reconstruction target. As these features exhibit high intraobject homogeneity, this provides a stronger supervisory signal for object-centric representation learning on complex real-world data.

Fig. 3 illustrates the iterative update process of slot attention in feature space. Each point represents a token-wise feature vector extracted from the encoder, specifically, self-supervised features in the case of DINOSAUR [31], or CNN features in the case of the original slot attention [30]. At initialization, slot vectors are randomly sampled from a learnable Gaussian distribution and thus reside at arbitrary positions in feature space, independent of any cluster. Over R iterations, each slot competitively aggregates tokens with similar features via the attention mechanism and is progressively refined to represent a coherent group of tokens via the GRU. Upon convergence, each slot vector moves toward the centroid of a semantically coherent token group, with the corresponding attention map $\mathcal{A}$ capturing the spatial localization of that group. This clustering behavior naturally leads to object-level decomposition, as tokens belonging to the same object tend to share semantically similar features, causing them to group together and be captured by a single slot.

B. Slot Attention in Optical Imagery

This section presents experimental results of applying DINOSAUR, based on DINOv2 [52], to real-world optical imagery. Fig. 4(1) visualizes the feature vectors extracted from DINOv2 using uniform manifold approximation and projection (UMAP) [53], alongside the corresponding slot attention outputs. In the UMAP visualization, each point represents a patch-level feature vector, with colors indicating ground truth (GT) object labels (background, cat, and dog).

The visualization reveals that feature vectors belonging to different object regions exhibit a clear semantic grouping tendency in the feature space. This suggests that DINOv2 [52], pretrained on large-scale data, has learned representations that cluster semantically similar regions within the feature space. Such features provide favorable conditions for the competitive mechanism of slot attention to aggregate semantically coherent tokens into a single slot. As a result, the slot attention outputs closely approximate object-level segmentation of the input image, showing that effective object-level decomposition can be achieved in real-world optical imagery.

C. Slot Attention in SAR Imagery

However, SAR imagery presents fundamentally different characteristics from optical imagery. In optical imagery, strong spatial continuity and rich visual cues such as color and texture encourage regions belonging to the same object to naturally cluster together in feature space. In contrast, SAR imagery is formed through the scattering properties of electromagnetic waves and is inherently composed of a collection of scattering points, resulting in relatively weak spatial continuity and semantic cohesion. Furthermore, the absence of color information and the presence of speckle noise limit the availability of intuitive textural cues. As a result, the visual feature space exhibits limited target–clutter separability, making it difficult to form object-specific feature homogeneity.

To analyze this, we extract feature vectors from SAR imagery and visualize them using UMAP, following the same procedure as in the optical imagery experiments. Specifically, we compare features extracted from SARATRX [12], a SAR-domain foundation model, and DINOv2 [52] pretrained on the ATRNet-STAR dataset [9]. Fig. 4(2) presents the UMAP visualizations of both models alongside their corresponding slot attention maps.

The visualizations reveal that even with an SAR foundation model, feature vectors corresponding to targets and clutter frequently exhibit intermixed distributions in feature space, without the clear separation observed in optical imagery. In such conditions, the features referenced during slot updates in slot attention are insufficient to discriminate between targets and clutter, resulting in entanglement where targets and clutter are collapsed into a single slot. These observations highlight the importance of refining the features referenced during slot updates to obtain more discriminative object representations.

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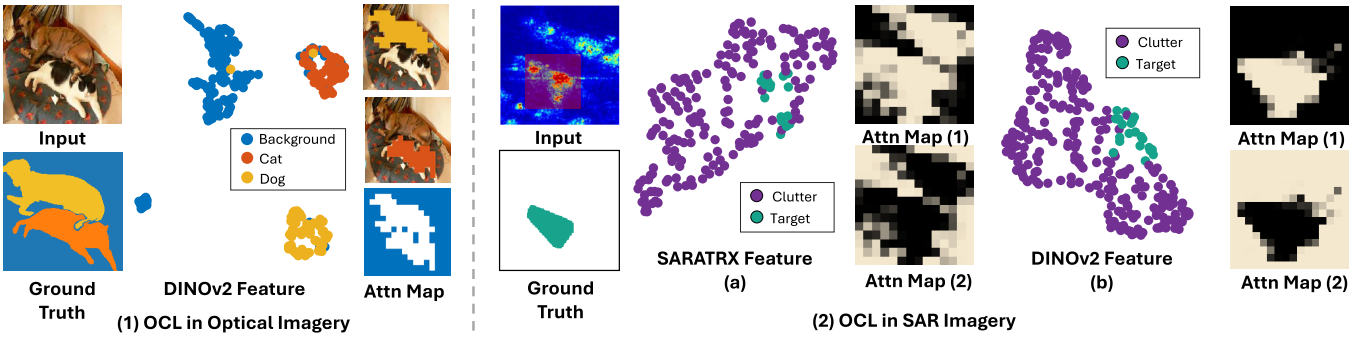


Fig. 4. Comparison of slot attention in optical and SAR imagery: 1) in optical imagery, DINOv2 features form distinct semantic clusters, enabling slot attention to achieve object-level decomposition and 2) in SAR imagery, both (a) SARATRX and (b) DINOv2 features show intermixed target–clutter distributions in feature space, causing slot attention to collapse targets and clutter into a single slot rather than producing distinct decomposition.

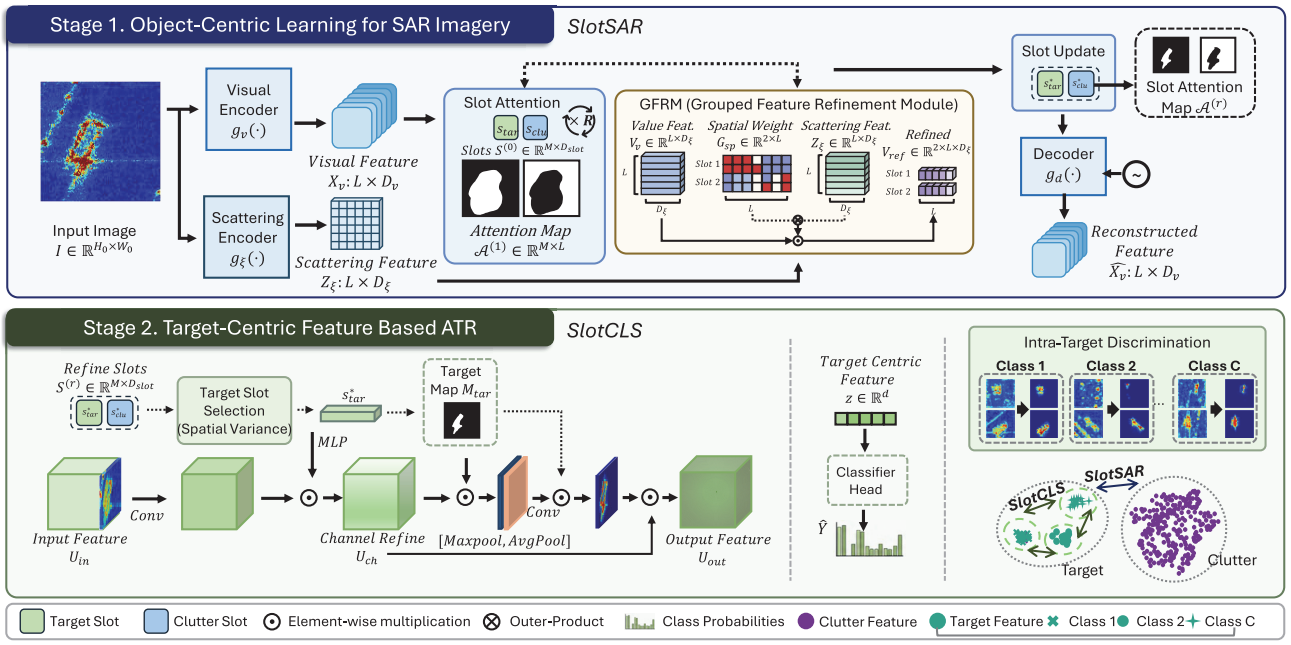


Fig. 5. *Overall pipeline of the proposed SlotATR framework.* Stage 1 (SlotSAR) takes an input SAR image and extracts visual and scattering features via dual encoders. The GFRM refines the grouping features by integrating spatial weights and scattering information, enabling slot attention to group recurring visual and scattering patterns into object-centric representations without mask annotations. Stage 2 (SlotCLS) selects the target slot and applies it to the input feature map through channel and spatial attention, producing target-centric features for discriminative classification.

III. Methodology

To reduce the dependence on hyperparameters in preprocessing and the influence of target mask annotations during FLCR, we propose SlotATR, an OCL-based SAR ATR framework. The overall pipeline is illustrated in Fig. 5.

SlotATR consists of two stages. In the first stage, SlotSAR is trained to decompose SAR imagery into slot-wise representations, separating target and clutter regions and obtaining discriminative feature representations for each. In the second stage, the target slot representation disentangled by SlotSAR is incorporated into SlotCLS, enabling target-centric classification without requiring target mask annotations. SlotCLS is designed to be flexibly integrated with various ATR backbones, allowing seamless incorporation into existing ATR pipelines.

A. SlotSAR

1) Motivation: The similarity between target and clutter feature distributions in SAR imagery weakens slot competition during object grouping. Consequently, relying on visual features makes it difficult for slots to converge to distinct object representations.

While developing an SAR foundation model that inherently produces discriminative features between targets and clutter could address this issue, it requires large-scale SAR data and substantial computational cost, which falls outside the scope of this work. Therefore, we instead adopt the approach of refining the features referenced during slot updates, which can be applied without retraining the existing pretrained SAR backbone and operates at the OCL module level, making it practically applicable to pretrained SAR backbones.

To guide this refinement, we focus on exploiting the intrinsic physical properties of SAR signals, rather than relying on visual inductive biases from optical-based OCL such as color or texture. As illustrated in Fig. 2, targets, which are typically metallic structures, tend to exhibit relatively consistent scattering responses and their spatial distributions across vary-

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ing background environments, such as industrial facilities, forested areas, or urban backgrounds [39], [40]. In contrast, clutter scattering responses and their spatial distributions tend to exhibit variability depending on material composition and environmental context [9]. Based on these observations, we hypothesize the following tendency in variability across sensing environments: $\mathrm{Var}_b[\mathbb{E}(z \mid \mathcal{R}_{\mathrm{tar}}, b)] < \mathrm{Var}_b[\mathbb{E}(z \mid \mathcal{R}_{\mathrm{clu}}, b)]$, where $\mathcal{R}_{\mathrm{tar}}$ and $\mathcal{R}_{\mathrm{clu}}$ denote the target and clutter regions, respectively; b denotes background or environmental conditions such as terrain type and surrounding structures; z denotes the feature representation; and $\mathbb{E}(\cdot \mid \cdot)$ and $\mathrm{Var}_b[\cdot]$ denote the conditional expectation and variance with respect to background variation. Accordingly, motivated by this variability asymmetry, we aim to learn the corresponding object-centric representations (a target slot s_{tar} and a clutter slot s_{clu}) through iterative slot updates.

2) Architecture: SlotSAR takes an input image and produces object-centric slot representations $s_{\mathrm{tar}}, s_{\mathrm{clu}} \in \mathbb{R}^{D_{\mathrm{slot}}}$ and their corresponding spatial attention maps $M_{\mathrm{tar}}, M_{\mathrm{clu}} \in \mathbb{R}^{H_f \times W_f}$, capturing the feature and localization information of the target and clutter regions, respectively. To this end, SlotSAR consists of three stages: feature encoding, GFRM, and slot attention, where GFRM is interposed between the feature encoding and slot attention stages to refine the features prior to slot updates.

Feature Encoding: Given an input SAR image $I \in \mathbb{R}^{H_0 \times W_0}$, SlotSAR extracts two complementary features: scattering features from a wavelet scattering network (WSN) [37] and visual features from an SAR foundation model. The scattering and visual features provide enhanced local scattering responses with improved signal-to-noise ratio (SNR) and high-level semantic information, respectively, providing complementary information for subsequent slot updates.

a) Scattering Feature: To capture local scattering responses and their spatial distributions in SAR imagery, we extract scattering features using a WSN. Conventional approaches, such as peak detection or SIFT [24], can be used to identify salient scattering points. However, these methods are sensitive to hyperparameter choices and rely on discrete or binarized representations, which limit stable gradient propagation. Therefore, we adopt a WSN to extract continuous scattering features.

The WSN, based on the wavelet transform, first measures the θ-directional component of the spatial frequency corresponding to scale a at position $\mathbf{b}$ in the image as follows:

$$W_I(a, \mathbf{b}, \theta) = \int I(\mathbf{x})\, \psi^*_{a,\theta}(\mathbf{x} - \mathbf{b})\, d\mathbf{x}. \tag{4}$$

Subsequently, an absolute value operation is applied to retain the magnitude of the wavelet responses while preventing positive–negative cancellation, and a Gaussian low-pass filter ϕ_J reduces sensitivity to local translations. The generalized mth order scattering coefficient is defined as $S_m = |\cdots|I * \psi_{\lambda_1}| * \psi_{\lambda_2} \cdots * \psi_{\lambda_m}| * \phi_J$. The WSN simultaneously measures scattering responses at each spatial location across multiple scales and orientations using a set of learnable filters, and extracts the final scattering features via pooling.

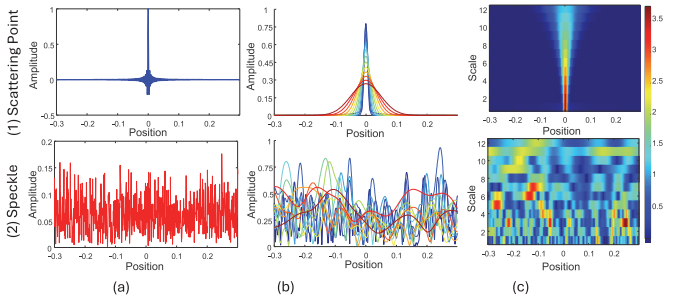


Fig. 6. Comparison of WSN responses between speckle and dominant scattering points: 1) dominant scattering point modeled as a sinc function following the SAR PSF and 2) speckle modeled as Rayleigh-distributed compound GN [15]. (a) Signal. (b) Multiscale response. (c) Scalogram.

In SAR imagery, man-made targets such as metallic structures consist of a small number of dominant scattering points formed by strong specular reflections. These scattering points appear spatially as sinc functions due to the point spread function (PSF) of the SAR system

$$s_{AC}(t_r, t_a) = A_0, \rho_r\left[t_r - \frac{2R_0}{C}\right]\rho_a(t_a), \quad e^{-j\frac{4\pi f_0 R_0}{C}} e^{j2\pi f_{t_a} t_a} \tag{5}$$

where A_0 is the scattering amplitude, $\rho_r(\cdot)$ and $\rho_a(t_a)$ are the range and azimuth impulse responses, respectively, R_0 is the closest range to the target, C is the speed of light, f_0 is the carrier frequency, and f_{t_a} is the Doppler frequency. The spatial response of a point target after range and azimuth compression takes the form of a sinc function. This sinc response, in both the range and azimuth directions, transforms in the frequency domain into a rect function with a uniform response within a finite bandwidth B_r and B_a

$$\begin{aligned} \rho_r(t_r) = \mathrm{sinc}(B_r t_r) &\xrightarrow{\mathcal{F}} \quad P_r(f) = \mathrm{rect}\left(\frac{f}{B_r}\right) \\ \rho_a(t_a) = \mathrm{sinc}(B_a t_a) &\xrightarrow{\mathcal{F}} \quad P_a(f) = \mathrm{rect}\left(\frac{f}{B_a}\right). \end{aligned} \tag{6}$$

This implies that dominant scattering points contain consistent frequency components within a finite bandwidth, enabling consistent responses across multiple wavelet scales [Fig. 6(1)].

In contrast, clutter and speckle consist of a large number of distributed weak scattering points that combine incoherently, forming a stochastic signal [15]. Due to this nature, matching with wavelet filters varies across scales: at some scales, responses arise from fluctuation components, while at others, they are attenuated by averaging or cancellation effects. As a result, the responses vary irregularly across scales without maintaining a consistent location or pattern [Fig. 6(2)]. To illustrate this, Fig. 6 visualizes the WSN responses by modeling dominant scattering points as a sinc function following the SAR PSF, and the clutter as Rayleigh-distributed speckle based on the compound Gaussian model [15]. The simulation shows that the sinc forms stable responses at consistent locations across multiple scales, whereas the speckle exhibits irregularly fluctuating response patterns.

Consequently, dominant scattering points exhibit low fluctuation and consistent responses across filters of multiple scales and orientations, whereas clutter shows high fluctuation and irregular responses. During pooling, these consistent responses

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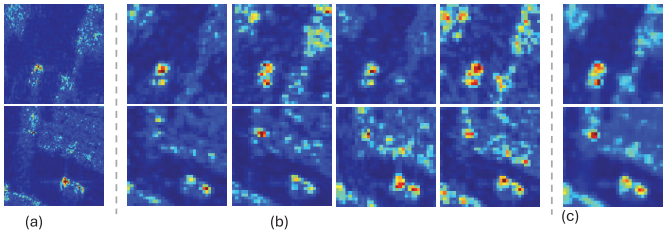


Fig. 7. Visualization of WSN processing on SAR images. (a) Input SAR images. (b) WSN responses across multiple scales and orientations before pooling. (c) Aggregated feature representation after pooling.

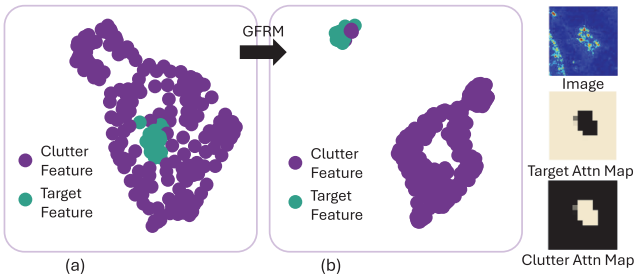


Fig. 8. Effect of GFRM on feature distributions and slot attention maps. (a) Initial feature distributions. (b) Refined feature distributions after GFRM.

are reinforced, while irregular clutter responses are relatively reduced, resulting in improved SNR. This characteristic is also consistently observed in SAR imagery (Fig. 7). Consequently, the resulting WSN features represent local scattering responses and their spatial distributions, which are subsequently used for feature refinement during slot updates.

In the WSN g_ξ, the wavelet filter parameters (orientation, Gaussian scale, center frequency, and aspect ratio) are jointly optimized during training [37]. This allows the filters to adaptively capture the scattering characteristics of SAR imagery beyond fixed Morlet wavelets. The extracted features $X_\xi = g_\xi(I)$ are subsequently processed through a lightweight encoder and self-attention module to yield $Z_\xi \in \mathbb{R}^{L\times D_\xi}$. These scattering features serve as a discriminative embedding within the GFRM to refine value features during slot updates.

b) Visual Feature: Although WSN extracts enhanced scattering responses with improved SNR, it lacks high-level semantic understanding, making it difficult to capture object-level relationships. As a result, relying on scattering features leads to spatially diffuse and unstable attention [31]. Therefore, we incorporate semantic features that provide global contextual cues and coarse spatial guidance. We extract visual features from an SAR foundation model g_v as $X_v = g_v(I)$, $X_v \in \mathbb{R}^{L\times D_v}$, where D_v denotes the feature dimensionality. The SAR foundation model is pretrained using self-supervised learning, enabling it to capture high-level representations of SAR imagery.

Based on X_v, initial slots are obtained, yielding slot representations $S \in \mathbb{R}^{M\times D_v}$ corresponding to the target slot s_{tar} and clutter slot s_{clu}. The initial attention map $\mathcal{A}^{(1)} \in \mathbb{R}^{M\times L}$ is derived from the first iteration of slot attention, providing coarse positional information of target candidates. Although not fully disentangled, it emphasizes high-response regions via similarity-based attention, serving as a reliable prior for guiding feature refinement in GFRM (Fig. 2).

GFRM: As discussed above, relying solely on visual features makes it difficult for slots to converge to distinct object representations. To complement visual features for object grouping, we introduce the GFRM to refine the value features prior to slot updates. Since slot attention updates each slot by aggregating value features weighted by attention, GFRM transforms the original value feature $V_v \in \mathbb{R}^{L\times D_\xi}$ into a refined representation V_{ref}.

To this end, GFRM exploits two complementary components: a spatial weight G_{sp}, derived from the initial attention map $\mathcal{A}^{(1)} \in \mathbb{R}^{M\times L}$, and a scattering feature Z_ξ^{slot}, extracted during feature encoding. Since the initial attention map $\mathcal{A}^{(1)}$ is computed solely from visual features before GFRM refinement, it may incompletely capture target regions in clutter-dominant environments. To alleviate this issue, the spatial weight G_{sp} adopts a soft-weighting strategy instead of hard masking and is defined as $G_{\text{sp}} = 1 + (\mathcal{A}^{(1)} - \mu_{\mathcal{A}})$, where $\mu_{\mathcal{A}}$ denotes the mean of the attention map. Unlike hard masking, soft weighting only attenuates low-attention responses rather than removing them completely. Consequently, target regions receiving insufficient attention during the initial iteration are partially preserved, allowing them to be progressively reinforced through subsequent slot updates.

GFRM combines the two components to refine the value feature as $V_{\text{ref}} = (Z_\xi^{\text{slot}} \otimes G_{\text{sp}}) \odot V_v$, where $\otimes$ and $\odot$ denote the outer product and element-wise multiplication, respectively. This refinement incorporates local scattering responses and their spatial distributions into the visual features, revealing additional feature variations between target and clutter regions that are weakly expressed by visual features alone.

Spatially irrelevant responses are attenuated by G_{sp} based on the initial attention map. Within the retained regions, enhanced local scattering responses and their spatial distributions complement the visual features for object-level grouping. Consequently, GFRM enriches the value features with local scattering responses and their spatial distributions. As shown in Fig. 8, the refined value features exhibit reduced overlap between target and clutter distributions, providing a more informative representation for subsequent object grouping.

Slot Attention: The refined V_{ref} is then referenced during slot updates, replacing the value features in the standard slot update

$$\tilde{S}^{(r)} = \text{GRU}\left(S^{(r-1)},\ \left(\bar{\mathcal{A}}^{(r)}\right)^{\top} \cdot V_{\text{ref}}\right) \tag{7}$$

$$S^{(r)} = \tilde{S}^{(r)} + \text{MLP}\left(\text{LayerNorm}\left(\tilde{S}^{(r)}\right)\right) \tag{8}$$

where $\bar{\mathcal{A}}_{i,j}^{(r)} = \mathcal{A}_{i,j}^{(r)} / \sum_{m=1}^{L} \mathcal{A}_{m,j}^{(r)}$ is the row-normalized attention map, and the remaining slot update procedure follows standard slot attention.

Through iterative slot updates, visual and scattering patterns in V_{ref} are progressively aggregated into their corresponding slots. As shown in Fig. 2, target regions tend to exhibit relatively consistent dominant scattering responses and spatial distributions across different background environments, whereas the scattering intensity and spatial distributions of clutter tend to vary with the surrounding environment. During

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training, slot attention utilizes the refined value features and progressively groups recurring target-related visual and scattering patterns through iterative slot updates, thereby reinforcing target-related intraobject homogeneity. Consequently, after R iterations, s^*_{tar} and s^*_{clu} represent the target and clutter regions, respectively.

3) Training: The refined target and clutter slots s^*_{tar}, s^*_{clu} are passed through a lightweight MLP decoder $g_d(\cdot)$. The reconstruction loss encourages the decoded output $\hat{X}_v$ to match the visual features X_v extracted from the encoder

$$\mathcal{L}_{\text{Recon}} = \left\| X_v - g_d\left(s^*_{\text{tar}},\ s^*_{\text{clu}}\right) \right\|_2 \tag{9}$$

which has been shown to provide more robust training signals for real-world datasets [31].

B. SlotCLS

While SlotSAR focuses on securing discriminability between targets and clutter, SlotCLS aims to reduce clutter-induced interference while preserving target-discriminative responses. In ATR, interclass discriminability stems from rich spatial features encoding the fine-grained geometric structures and scattering patterns of targets. However, since slot representations in SlotSAR are learned to summarize the entire scene for target/clutter separation, they are not well-suited to preserve such fine-grained spatial features. Therefore, we propose SlotCLS, a module that injects target localization and semantic features obtained from OCL into the spatial feature extraction process of existing ATR backbones by coupling the outputs of SlotSAR with intermediate feature maps.

SlotCLS takes as input an intermediate feature map $U_{\text{in}} \in \mathbb{R}^{C_f \times H_f \times W_f}$ from the ATR backbone, along with the spatial attention maps $M_{\text{tar}}, M_{\text{clu}}$ and slot vectors $s^*_{\text{tar}}, s^*_{\text{clu}}$ from SlotSAR. It refines the feature map by integrating target-aware semantic information from s^*_{tar} and spatial information from M_{tar} through channel-wise and spatial modulation. This process enhances target-discriminative features while reducing clutter-induced interference, enabling target-centric ATR without mask annotations. The components of SlotCLS are described as follows: target selector, channel attention, and spatial attention.

Target Selector: Since slot attention produces permutation-equivariant outputs, the slot index corresponding to the target varies arbitrarily across inputs. Explicitly identifying the target slot is therefore necessary. A naive approach of selecting the slot with the minimum attention value is prone to misidentification when clutter occupies a large spatial region. Therefore, we select the target slot based on the spatial compactness of each attention map. Under the single-target assumption, the target mask is denoted as $B_{\text{tar}} \in \{0,1\}^{H_f \times W_f}$, and the clutter region is defined as its complement $1 - B_{\text{tar}}$.

In SAR imagery, the target is composed of a collection of scatterers and forms a spatially compact cluster, whereas clutter is distributed broadly across the remaining image area. As illustrated in Fig. 9, this structural difference is reflected in the attention maps, where targets appear concentrated and clutter more dispersed. Therefore, we use spatial variance as a selection criterion and identify the target slot as the one with the minimum variance.

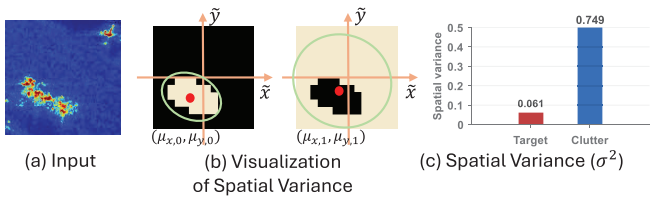


Fig. 9. Visualization of spatial variance for target slot selection. (a) Input SAR image. (b) Target and clutter attention maps exhibit distinct spatial variance, where the target region is spatially compact and the clutter region is broadly distributed. (c) Quantitative comparison of spatial variance confirms that the target slot consistently yields lower variance than the clutter slot.

Concretely, the attention map $M_m \in \{0,1\}^{H_f \times W_f}$ of the mth slot is normalized into a probability distribution, with pixel coordinates normalized to $[-1, 1]$

$$\pi_m(\tilde{x}, \tilde{y}) = \frac{M_m(\tilde{x}, \tilde{y})}{\sum_{\tilde{x}', \tilde{y}'} M_m(\tilde{x}', \tilde{y}')}. \tag{10}$$

The spatial variance σ^2_m is then defined as

$$\sigma^2_m = \sum_{\tilde{x}, \tilde{y}} \left[\left(\tilde{x} - \mu_{x,m}\right)^2 + \left(\tilde{y} - \mu_{y,m}\right)^2 \right] \pi_m(\tilde{x}, \tilde{y}). \tag{11}$$

$m_{\text{tar}} = \arg\min_m \sigma^2_m$, where $(\mu_{x,m}, \mu_{y,m})$ denotes the center of mass of π_m. Spatial variance is computed with respect to the center of mass of each mask, making it invariant to spatial translation and sensitive to the spread of the activation rather than its absolute location. This property makes the criterion robust to variations in target position and scale.

Channel Attention: The selected target slot vector s^*_{tar} is used to generate channel-wise attention weights via an MLP to modulate the channel-wise importance of the feature map. Since the slot vector encodes the semantic attributes of the target in a channel-wise manner, utilizing it as channel attention enables selective emphasis on channels relevant to the target. The channel attention map U_{ch} is computed as

$$U_1 = f^{3\times 3}(U_{\text{in}}) \tag{12}$$

$$U_{\text{ch}} = U_1 \odot \left(1 + \text{MLP}\left(s^*_{\text{tar}}\right)\right) \tag{13}$$

where $f^{3\times 3}(\cdot)$ denotes a 3×3 convolution, and $(1 + \text{MLP}(s^*_{\text{tar}}))$ serves as a residual channel attention weight derived from the target slot vector.

Spatial Attention: Given the channel refinement map U_{ch}, the target localization map M_{tar} inferred from the selected target slot attention map is used as spatial attention to highlight the target region. First, M_{tar} is multiplied element-wise with U_{ch} to produce a masked feature map U_{mask} that emphasizes the target region: $U_{\text{mask}} = U_{\text{ch}} \odot M_{\text{tar}}$. Average pooling and max pooling are then applied to U_{mask} along the channel dimension, and the results are concatenated and passed through a 1×1 convolution to generate the spatial attention map R_{sp}

$$R_{\text{sp}} = f_{1\times 1}\left(\text{Concat}\left(\text{AvgPool}(U_{\text{mask}}),\ \text{MaxPool}(U_{\text{mask}})\right)\right). \tag{14}$$

Finally, the spatial attention map is obtained by combining the target localization map and the learned spatial response as $G_{\text{cls}} = M_{\text{tar}} \odot R_{\text{sp}}$. The output feature map is then computed as

$$U_{\text{out}} = U_{\text{ch}} \odot G_{\text{cls}}. \tag{15}$$

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The refined U_{out} is propagated to subsequent backbone layers and optimized with a cross-entropy loss, enabling target-centric feature learning without mask supervision. During training, the weights of SlotSAR are frozen, and only SlotCLS is optimized. As a lightweight module, SlotCLS can be flexibly integrated with various ATR backbones, as experimentally validated in Section IV-D.

IV. Experimental Results

A. Dataset Description

In this study, we aim to mitigate clutter interference in SAR ATR while reducing reliance on ground-truth target masks. We therefore focus on the single-target classification setting widely adopted in the SAR ATR literature [9], [16], [17], [18], [20], [47], [54]. This setup assumes a postdetection scenario in which a single target is present within each image. Such an assumption is shared across many deep learning-based ATR models and provides a consistent baseline for comparison. Following this setting, we use two slots corresponding to the target and background clutter. Accordingly, we evaluate the proposed SlotATR using two representative SAR ATR benchmarks: MSTAR [54] and ATRNet-STAR [9]. The detailed dataset configurations are summarized in Table I.

1) The MSTAR dataset contains ten target classes acquired over grass surfaces, resulting in relatively simple scenes dominated by natural clutter. Owing to its controlled acquisition conditions and homogeneous background, MSTAR has long served as a standard benchmark in SAR ATR research. Using MSTAR, we examine whether SlotSAR can reliably disentangle targets from background even in clean and structured environments, and whether its object-centric feature preserves discriminative target representations under minimal clutter.
2) In contrast, ATRNet-STAR is a large-scale benchmark comprising 40 vehicle classes collected under diverse Ku/X-band sensing conditions, including variations in azimuth, depression angle, and target position. More importantly, it spans heterogeneous environments such as woodland, city, and factory scenes, introducing a mixture of natural and artificial clutter that closely reflects real EOCs. These characteristics make ATRNet-STAR one of the most challenging datasets for SAR ATR. Experiments on ATRNet-STAR are conducted to assess whether SlotSAR can effectively separate targets from complex and highly heterogeneous clutter, and whether SlotCLS can maintain discriminative and stable target representations for accurate classification under severe EOCs.

TABLE I
Dataset Configurations of MSTAR [54] and ATRNet-STAR [9]

Dataset	Setting	Set	# Types	# Img.	
				Ground Range	Slant Range
MSTAR	SOC	Train	10	2,747	
		Test	10	2,425	
ATRNet-STAR	SOC-40	Train	40	68,091	67,780
		Test	40	29,284	29,169
	SOC-50	Train	50	18,071	18,071
		Test	50	17,603	17,613
	EOC-Azimuth	Train	40	18,636	18,592
		Test	40	78,739	78,357
	EOC-Band	Train	40	27,711	27,653
		Test	40	27,763	27,732
	EOC-Depression	Train	40	24,361	22,206
		Test	40	73,014	74,743
	EOC-Polarization	Train	40	24,361	24,246
		Test	40	73,014	72,703
	EOC-Scene	Train	40	19,584	19,584
		Test	40	77,791	77,365

B. Experimental Setup

1) Evaluation of OCL: Implementation Details: For OCL evaluation, we use the MSTAR and ATRNet-STAR datasets. For ATRNet-STAR, the official train/test splits provided for each benchmark setting, as summarized in Table I, are adopted, and ground-range images are used for OCL training. Since ATRNet-STAR already contains sufficient variations in target position and scale, the original images are used without additional data augmentation during OCL training. For MSTAR, the SOC setting is followed, where images acquired at a depression angle of 17° are used for training and those at 15° are used for testing. As the MSTAR dataset provides limited variations in target position, target scale, and background clutter [9], data augmentation is applied to improve the diversity of the training data. Specifically, each training image is randomly translated by up to 60% of the image size using reflection padding, followed by gamma contrast augmentation with a gamma value randomly sampled from the range $[0.5, 3.0]$, and random crop-resize augmentation with a crop scale randomly selected from $[0.55, 1.0]$. The MSTAR training set is augmented to five times its original size. For both datasets, 10% of the training set is randomly held out as a validation set.

The scattering encoder is implemented based on the parametric scattering network [37]. The scattering scale is set to $J = 2$ to define the receptive field of the wavelet filters. Second-order scattering coefficients are used throughout all experiments, and the effect of the scattering order is analyzed in the ablation study (Section IV-C3). Tight-frame initialization is adopted to ensure stable optimization and uniform frequency-band coverage during training. A separate learning rate of 0.1 is assigned to the orientation, Gaussian scale, frequency, and aspect ratio parameters of the scattering encoder.

SlotSAR is optimized using the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with an initial learning rate of 1×10^{-4} and is trained for 200 epochs on all datasets. The learning rate is linearly warmed up over the first 10k steps and subsequently decayed using an exponential schedule. Gradient clipping with a threshold of 1 is applied to stabilize training. The batch size is set to 64. The slot dimension is set to $D_{\text{slot}} = 256$, and $R = 3$ refinement iterations are performed in slot attention. The feature token dimensions are set to $D_\xi = 64$, $L = 196$, and $D_v = 512$. The decoder is implemented as an MLP.

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Evaluation Metrics: We evaluate the OCL performance of SlotSAR using adjusted rand index (ARI), mean best overlap (mBO), mean intersection over union (mIoU), and $F1$-score. ARI measures clustering consistency, while mBO quantifies the overlap between predicted target regions and ground-truth masks. mIoU measures the average intersection over union (IoU) of the target and background regions, and $F1$-score, defined as the harmonic mean of precision and recall, measures segmentation quality. In the ablation experiments, binary masks derived from attention maps are compared against annotations to evaluate whether each slot captures its corresponding target region. Following prior work [30], [31], Hungarian matching is applied to align predicted slots with ground-truth regions. For qualitative analysis, we project high-dimensional feature maps into a 2-D space using principal component analysis (PCA) [52]. In all quantitative comparisons, the best and second-best results are highlighted in bold and underlined, respectively.

Ground-Truth Mask Construction: To quantitatively evaluate target–clutter separation, we construct ground-truth masks for both MSTAR and ATRNet-STAR datasets following prior works [16], [55]. For ATRNet-STAR, bounding box annotations are directly used, while for MSTAR, a fixed central bounding box is applied. Within each box, target regions are extracted using high-intensity pixels, based on the observation that strong scattering responses correspond to target structures [16], [20], [21]. Morphological filtering, convex hull, and filling are applied to obtain compact masks. Finally, all masks are visually inspected and manually refined when necessary. Although this process relies on location and intensity heuristics, SlotSAR performs disentanglement without such annotations. All comparative methods are implemented under the same experimental settings to ensure a fair comparison.

2) Evaluation of ATR: Implementation Details: ATR performance is evaluated using the EOC-S setting of ATRNet-STAR to assess generalization under diverse clutter environments. In the EOC-S setting, the training set consists of the relatively low-clutter bare soil and sandstone scenes, whereas the test set contains the more challenging city, factory, and woodland scenes with significantly stronger clutter interference [9], [22]. This setting evaluates whether a model trained under limited clutter conditions can reliably recognize targets in previously unseen environments with severe clutter interference, making it well suited for the objective of clutter mitigation addressed in this work.

Comparisons are conducted with SAR ATR methods from different research directions, including pixel-level clutter reduction (ESENet [62]), an SAR foundation model (SARATRX [12]), FLCR (HDANet [17] and CRM [16]), background-robust learning (BRL [26]), attributed scattering center (ASC [23])-based methods (SANet [44] and ASC-SepNet [63]), and ASC-based graph methods (VSFA [47], LDSF [64], and SARGTR [22]). Since ASC-based methods require phase information to extract scattering centers, they are trained and evaluated using slant-range images. For the MSTAR dataset, experiments follow the SOC setting summarized in Table I. For both datasets, 10% of the training set is randomly selected as the validation set. The models are trained without data augmentation. SlotCLS is integrated into the feature extraction stage of each backbone. All models are trained using the Adam optimizer with an initial learning rate of 5×10^{-4} and a batch size of 32 for 50 epochs, followed by an additional 40 epochs with a decayed learning rate. SlotSAR used by SlotCLS is trained separately for each experimental setting.

Evaluation Metrics: Classification performance is primarily evaluated using overall accuracy. In addition, precision and recall are reported in Section IV-D5 to assess prediction bias across different target sizes.

C. Evaluation of OCL

1) Feature Separability Analysis With GFRM: GFRM incorporates local scattering responses and their spatial distributions into the visual features to facilitate object grouping during slot updates. To evaluate which auxiliary features are effective for this refinement, we compare scattering-based features (WSN [37], peak detection [58], and SIFT [59]) with hand-crafted features (HOG [56] and Canny [57]) that capture edge and gradient information. Furthermore, using WSN as the auxiliary feature, we compare GFRM with conventional feature integration strategies, including encoder-level and slot-level fusion.

As shown in Table II, the baseline OCL model using SARATRX [12] as the visual encoder still exhibits substantial overlap between target and clutter features within the attended regions. To quantitatively analyze the feature distributions, we measure interclass dispersion within the attention-highlighted area as $\sigma(F \odot \mathcal{A}^{(1)})$, where σ denotes the standard deviation, and intraclass compactness over target-only pixels as $\sigma(F \odot G_{gt})$. The former measures the dispersion between target and clutter features within the region involved in slot updates, while the latter measures the homogeneity of target representations. Table II summarizes these metrics together with OCL performance across different auxiliary features and integration strategies. For DINOSAUR, F corresponds to the visual feature V_v, whereas for auxiliary features, F denotes $Z_{\xi}^{\text{slot}} \otimes G_{\text{sp}}$ obtained through GFRM.

From the perspective of auxiliary feature selection, different feature types exhibit distinct behaviors in interclass dispersion and intraclass dispersion. Compared with edge-based features such as HOG and Canny, scattering-based representations consistently exhibit higher interclass dispersion and lower intraclass dispersion, suggesting that scattering characteristics are more suitable for refining the value representation for object grouping. Among the scattering-based features, peak-based representations and SIFT exhibit lower interclass dispersion and higher intraclass dispersion than WSN. In contrast, WSN achieves the highest interclass dispersion $\sigma(F \odot \mathcal{A}^{(1)})$ and the lowest intraclass dispersion $\sigma(F \odot G_{gt})$, together with the highest mBO and ARI performance, supporting its effectiveness as an auxiliary feature for GFRM.

From the perspective of feature integration, conventional fusion-based approaches show limited effectiveness in SAR imagery. Encoder-level fusion via cross-attention degrades performance, possibly because early feature mixing may disturb the semantic structure of visual representations used for object

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TABLE II

COMPARISON OF AUXILIARY FEATURES AND FUSION STRATEGIES IN TERMS OF OCL PERFORMANCE (MBO, ARI) AND TARGET–CLUTTER SEPARABILITY METRICS ($\sigma(F \odot \mathcal{A}^{(1)})$, $\sigma(F \odot G_{gt})$, RATIO). FOR DINOSAUR, F CORRESPONDS TO THE VISUAL FEATURE V_v, WHEREAS FOR AUXILIARY FEATURES, F DENOTES $Z_{\xi}^{\text{SLOT}} \otimes G_{sp}$ OBTAINED VIA THE GFRM

Category	Method	mBO [%] ↑	ARI [%] ↑	$\sigma(F \odot \mathcal{A}^{(1)})$ ↑	$\sigma(F \odot G_{gt})$ ↓	Ratio ↑
Baseline	DINOSAUR [31]	18.86	16.20	0.18	0.11	1.64
Auxiliary Features (X_ξ)	WSN [37]	**29.19**	**33.36**	**0.302**	**0.13**	**2.31**
	HOG [56]	17.83	13.07	0.237	0.351	0.68
	CANNY [57]	18.33	15.59	0.298	0.337	0.88
	Peak Detection [58]	20.64	21.41	0.295	0.304	0.97
	SIFT [59]	19.07	16.60	0.241	0.22	1.10
	Gaussian Noise	17.40	11.10	0.201	0.19	1.05
Encoder-level Fusion	Cross Attention [60]	7.87	1.19	0.027	0.14	0.19
Slot-level Fusion	SMSA [61]	6.33	0.28	-	-	-
Feature Refinement	GFRM	**29.19**	**33.36**	**0.302**	**0.13**	**2.31**

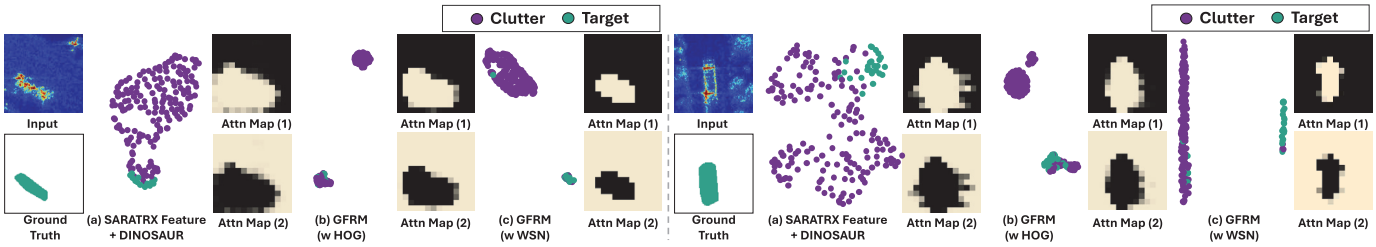


Fig. 10. UMAP visualization of refined features and corresponding slot attention maps for (a) DINOSAUR with SARATRX [31], (b) GFRM with HOG [56], and (c) GFRM with WSN [37]. Each point represents a patch token extracted from a single input image, where purple and green denote clutter and target tokens, respectively.

grouping. Slot-level fusion via SMSA [61] also yields limited performance, as the high similarity between target and clutter features may cause their mapping to similar codebook entries. In contrast, GFRM refines the value representation prior to slot updates rather than directly fusing heterogeneous features. This approach results in higher interclass dispersion, lower intraclass variation, and improved mBO and ARI scores. These results indicate that incorporating local scattering responses and their spatial distributions into the visual features provides a more effective feature space for object grouping.

To further examine how GFRM affects the feature distributions, we visualize the refined features using UMAP [53], along with the corresponding slot attention maps. As shown in Fig. 10, the baseline DINOSAUR (w SARATRX) features exhibit substantial overlap between target and clutter distributions, whereas the refined features after GFRM exhibit reduced overlap. In particular, WSN-based refinement shows less target–clutter overlap than HOG-based refinement, consistent with the quantitative results in Table II.

2) Comparison of OCL Methods: We compare the performance of SlotSAR with several OCL methods: slot attention [30], DINOSAUR [31], invariant slot attention (ISA) [65], and SMSA [61]. For fair comparison, we evaluate each method using both DINOv2 [52] and SARATRX, with a shared decoder architecture across all models. As shown in Table III, our method achieves superior performance across challenging operating conditions. Specifically, compared to the best-performing baselines, SlotSAR improves the average ARI, mIOU, mBO, and $F1$ Score by +7.83%, +3.01%, +5.62%, and +5.35%, respectively. The EOC-S setting poses a significant challenge, as the model is trained on simple clutter but evaluated in much more complex environments. These include woodland scenes with severe speckle noise from natural clutter, and industrial areas such as factories, where strong reflections arise from artificial clutter. As shown in Table IV, SlotSAR shows consistent performance despite these challenging conditions, achieving ARI improvements of +10.34% and +14.92% over DINOSAUR, and +5.17% and +8.34% over ISA in woodland and factory scenes, respectively. These results indicate effective target–clutter disentanglement in SAR imagery, even in complex unseen environments.

Fig. 11 presents a qualitative comparison between the baseline and the SlotSAR under various scenarios. To ensure a fair comparison, we visualize attention maps as binary masks for all models, allowing us to observe how well each slot focuses on the target. The evaluation covers a wide range of cases, from scenes with mild to severe natural clutter [Fig. 11(a)] and from low to high levels of artificial clutter [Fig. 11(b)] in the ATRNet-STAR dataset, as well as additional results from the MSTAR dataset [Fig. 11(c)]. From the results, we observe that: 1) under dominant natural clutter, SlotSAR achieves more precise localization and finer delineation of targets and 2) in complex scenes where man-made structures exhibit stronger backscatter than the target, it more effectively isolates target features from clutter. These results indicate that SlotSAR improves target–clutter disentanglement in the ATR setting under challenging environmental conditions.

3) Ablation Study: We conduct ablation studies on SlotSAR with respect to two aspects: the effect of the number of iterations R and the contribution of each component.

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TABLE III

QUANTITATIVE COMPARISON OF THE PROPOSED SLOTSAR FRAMEWORK WITH SIX METHODS ON ATRNET-STAR AND MSTAR DATASETS. EOC-A, EOC-B, EOC-D, EOC-P, AND EOC-S DENOTE AZIMUTH, FREQUENCY BAND, DEPRESSION ANGLE, POLARIZATION, AND SCENE, RESPECTIVELY

Datasets		ATRNet-STAR							MSTAR	ATRNet-STAR							MSTAR
		EOC-A	EOC-B	EOC-D	EOC-P	EOC-S	SOC-40	SOC-50	SOC	EOC-A	EOC-B	EOC-D	EOC-P	EOC-S	SOC-40	SOC-50	SOC
Methods \ Metric		Adjusted Rand Index (ARI) [%] ↑								mean Intersection over Union (mIOU) [%] ↑							
Slot Attention		4.46	22.91	12.32	17.31	1.29	0.79	16.58	10.23	40.92	52.20	47.23	50.10	32.44	31.46	51.08	49.17
DINOSAUR	DINOv2	0.39	0.35	0.40	0.40	0.28	0.50	3.43	0.23	46.97	47.29	47.05	47.00	46.81	46.93	47.26	47.37
	SARATRX	25.36	22.95	21.72	28.91	17.24	23.00	25.04	8.69	52.95	51.76	50.66	55.33	47.55	51.83	51.25	40.43
ISA	DINOv2	11.14	15.60	12.79	15.04	12.83	11.82	11.79	22.35	42.15	46.70	43.47	45.34	43.37	43.22	43.64	51.09
	SARATRX	34.87	36.51	38.81	36.26	25.65	36.86	26.82	35.80	60.22	60.93	62.12	60.36	54.77	60.27	55.56	59.51
SMSA	SARATRX	0.39	0.35	0.40	0.40	0.28	0.50	3.43	0.57	46.97	47.29	47.05	47.00	46.81	46.93	47.26	47.37
SlotSAR		**43.07**	**45.27**	**41.67**	**43.69**	**30.85**	**43.65**	**37.57**	**48.45**	**63.96**	**64.72**	**63.31**	**62.81**	**57.48**	**62.08**	**60.12**	**63.31**
Methods \ Metric		mean Best Overlap (mBO) [%] ↑								F1 Score [%] ↑							
Slot Attention		10.04	21.88	13.91	17.15	8.92	8.14	15.95	10.98	17.41	34.21	23.17	27.51	15.83	14.61	25.98	19.66
DINOSAUR	DINOv2	6.05	5.44	5.98	6.08	6.33	6.07	5.41	5.20	10.76	9.82	10.50	10.67	11.28	10.90	9.75	9.94
	SARATRX	24.51	22.25	22.04	26.82	19.89	22.77	23.47	14.27	37.25	34.91	34.29	40.03	31.03	35.83	34.60	24.96
ISA	DINOv2	16.25	17.71	17.08	18.43	17.51	16.51	15.44	21.93	26.11	28.94	27.15	28.82	27.89	26.92	25.57	35.73
	SARATRX	30.52	31.44	33.62	31.67	24.69	32.21	24.50	30.63	43.01	45.67	46.80	43.73	35.82	45.91	36.54	46.70
SMSA	SARATRX	6.05	5.44	5.98	6.08	6.33	6.07	5.41	5.19	10.76	9.82	10.50	10.67	11.28	10.90	9.75	9.94
SlotSAR		**37.07**	**37.58**	**35.21**	**37.01**	**28.47**	**36.81**	**32.10**	**40.01**	**50.21**	**51.97**	**49.52**	**48.96**	**38.86**	**49.11**	**43.38**	**54.94**

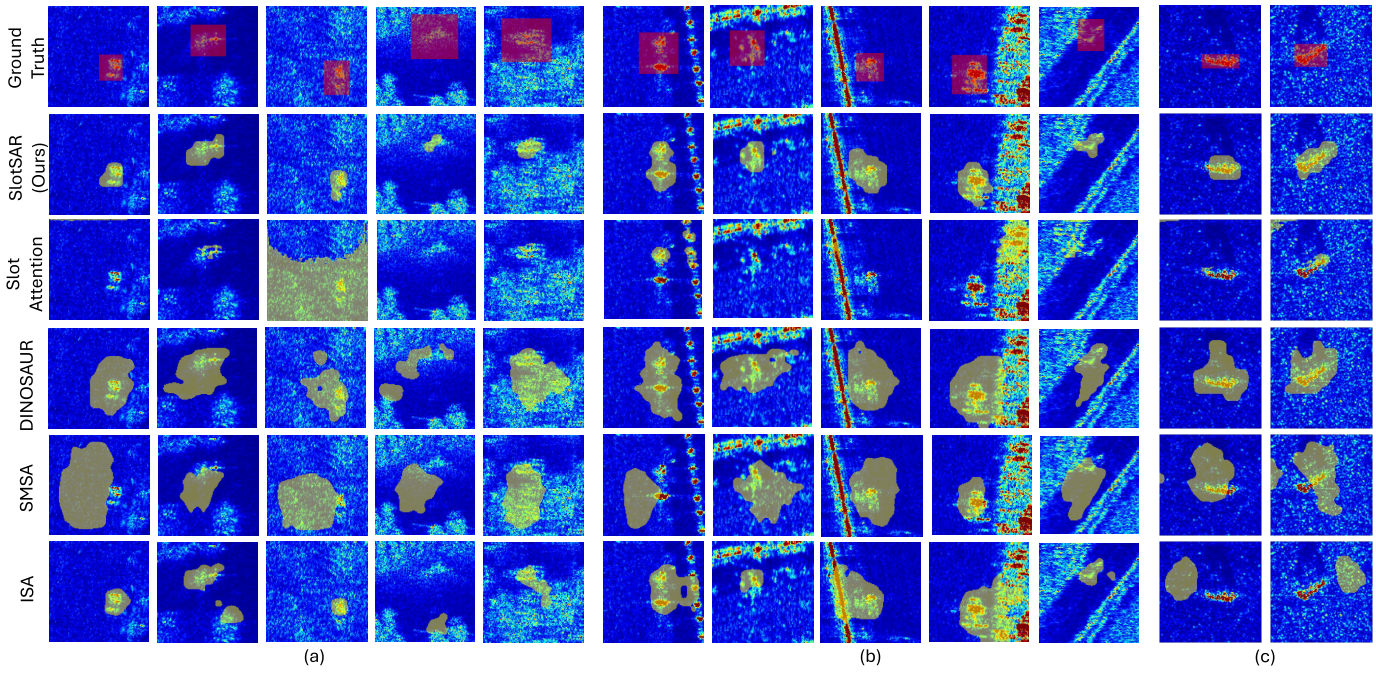


Fig. 11. Example results on the ATRNet-STAR and MSTAR datasets. (a) and (b) Results from ATRNet-STAR under natural clutter and artificial clutter conditions, respectively, while (c) results from the MSTAR dataset. The red regions indicate the ground-truth target regions, while the yellow target regions represent the areas predicted by the model.

Impact of Iteration in SA: We analyze the effect of R by varying the number of slot attention iterations. As shown in Fig. 12, three iterations achieve the best performance for SAR target–clutter disentanglement. As reported in the original slot attention work [30], using too few or too many iterations leads to performance degradation.

SlotSAR Components: Table V presents the ablation results of SlotSAR under various configurations, analyzing the contribution of each component. The first two rows show baselines using either visual or scattering features only. In 1), SlotSAR operates with the visual feature X_v, equivalent to DINOSAUR. In contrast, 2) replaces X_v with X_ξ, resulting in a significant performance drop. This indicates that the high-level semantic features provided by X_v are essential for SAR OCL.

Next, we analyze the contribution of each component within GFRM. Row (3) applies GFRM using only the spatial weight G_{sp}, which attenuates spatially irrelevant responses and yields a notable performance gain over the baseline. Row (4) applies

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TABLE IV
QUANTITATIVE COMPARISON OF ARI AND MBO UNDER EOC-S SETTING USING THE SARATRX VISUAL ENCODER

Metric	EOC-S	Slot Attention	DINOSAUR	ISA	SMSA	SlotSAR
ARI [%] ↑	City	0.57	14.16	23.71	0.33	**26.54**
	Factory	1.88	20.48	27.06	0.22	**35.40**
	Woodland	4.87	29.51	34.68	1.15	**39.85**
mBO [%] ↑	City	9.35	18.94	24.15	7.03	**26.03**
	Factory	8.39	20.96	24.83	5.72	**31.02**
	Woodland	9.23	25.65	29.34	4.28	**33.66**

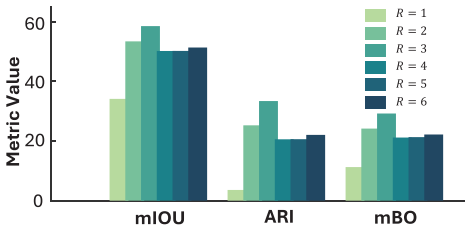


Fig. 12. Effect of the number of slot attention iterations r on OCL performance (mIOU, ARI, mBO).

TABLE V
ABLATION STUDY ON ENCODER AND GFRM COMPONENTS

Index	Encoder		GFRM		Metric	
	X_v	X_ξ	G_{sp}	Z_ξ	ARI [%] ↑	mBO [%] ↑
(1)	✓				16.20	18.86
(2)		✓			4.16	10.09
(3)	✓		✓		29.34	26.58
(4)	✓	✓		✓	31.16	27.46
(5)	✓	✓	✓	✓	**33.36**	**29.19**

GFRM using only Z_ξ, incorporating local scattering responses and their spatial distributions into the value representation for object grouping. Finally, row (5) combines both G_{sp} and Z_ξ, achieving the best ARI and mBO scores. These results indicate that the two components provide complementary information: G_{sp} spatially weights the value features based on the initial attention map, while Z_ξ complements the visual features with local scattering responses and their spatial distributions.

Impact of Scattering Order: We further investigate the effect of the scattering order in WSN on OCL performance and computational efficiency. As shown in Table VI, incorporating second-order scattering improves ARI and mBO from 27.62% to 24.73% and from 30.85% to 28.47%, respectively, indicating that capturing the spatial modulation of first-order scattering responses is beneficial for object grouping. Further increasing the scattering order to three yields only marginal improvements in ARI and mBO, while substantially reducing the processing speed from 17.11 to 5.66 FPS. These results suggest that second-order scattering provides an appropriate trade-off between OCL performance and computational efficiency.

4) Analysis of GFRM: We further analyze how the scattering information captured by WSN contributes to object grouping through GFRM and iterative slot updates. Specifically, we first examine whether WSN captures local scattering responses and their spatial distributions across target and clutter regions, and how GFRM incorporates these characteristics into the value representation for object grouping. We then investigate whether the refined value representation is effectively utilized during iterative slot updates. A qualitative comparison of the feature and slot attention maps across different models and iterations is presented in Fig. 13.

TABLE VI
EFFECT OF SCATTERING ORDER ON OCL PERFORMANCE EVALUATED USING PREDICTED MASKS AND COMPUTATIONAL EFFICIENCY

Order	ARI [%] ↑	mBO [%] ↑	FPS ↑
1	27.62	24.73	**63.71**
2	30.85	28.47	17.11
3	**31.37**	**29.89**	5.66

Scattering Energy and Target–Clutter Contrast: We first analyze the local scattering responses captured by WSN and how these responses are incorporated into the value representation through GFRM. Specifically, we examine the regional energy distributions of X_ξ across target and clutter regions, and then analyze how these responses are spatially weighted in $Z_\xi \otimes G_{sp}$. Since X_ξ encodes local scattering responses across multiple scales and orientations, its regional energy provides a quantitative measure of the scattering characteristics represented in each region.

Specifically, we compute the average feature energy of the target (E_{tar}) and clutter (E_{clu}) regions within the candidate area defined by $\mathcal{A}^{(1)}_{\text{tar}}$. This analysis is performed for both X_ξ and $Z_\xi \otimes G_{\text{sp}}$. We normalize these energies by their sum to account for variations in overall energy levels across scenes and further compute the ratio $E_{\text{tar}}/E_{\text{clu}}$ to quantify the relative scattering contrast between the two regions. In addition, we qualitatively compare X_ξ and $Z_\xi \otimes G_{\text{sp}}$ to examine how the local scattering responses and their spatial distributions are refined through GFRM. All experiments are conducted under the EOC-S setting.

Fig. 14 presents qualitative visualizations of X_ξ and $Z_\xi \otimes G_{\text{sp}}$. In scenes with natural clutter, X_ξ exhibits localized scattering responses around the target regions despite the presence of background clutter, illustrating the local scattering characteristics captured by WSN. After applying spatial weighting, $Z_\xi \otimes G_{\text{sp}}$ retains these responses while relatively attenuating spatially irrelevant responses. As a result, the scattering responses become more concentrated within the candidate regions. This observation shows how GFRM incorporates the local scattering responses captured by WSN and their spatial distributions into the value representation for subsequent object grouping.

Table VII reports the target and clutter energy measured within the candidate regions emphasized by M_{tar} for both X_ξ and $Z_\xi \otimes G_{\text{sp}}$. For X_ξ, the target–clutter energy contrast varies across environments. In woodland scenes dominated by natural clutter, target regions exhibit relatively stronger scattering responses, resulting in higher target–clutter contrast. In contrast, in city and factory scenes containing abundant artificial structures, target and clutter regions exhibit more similar scattering responses, resulting in reduced contrast. These observations show that WSN captures local scattering characteristics, while their target–clutter contrast varies

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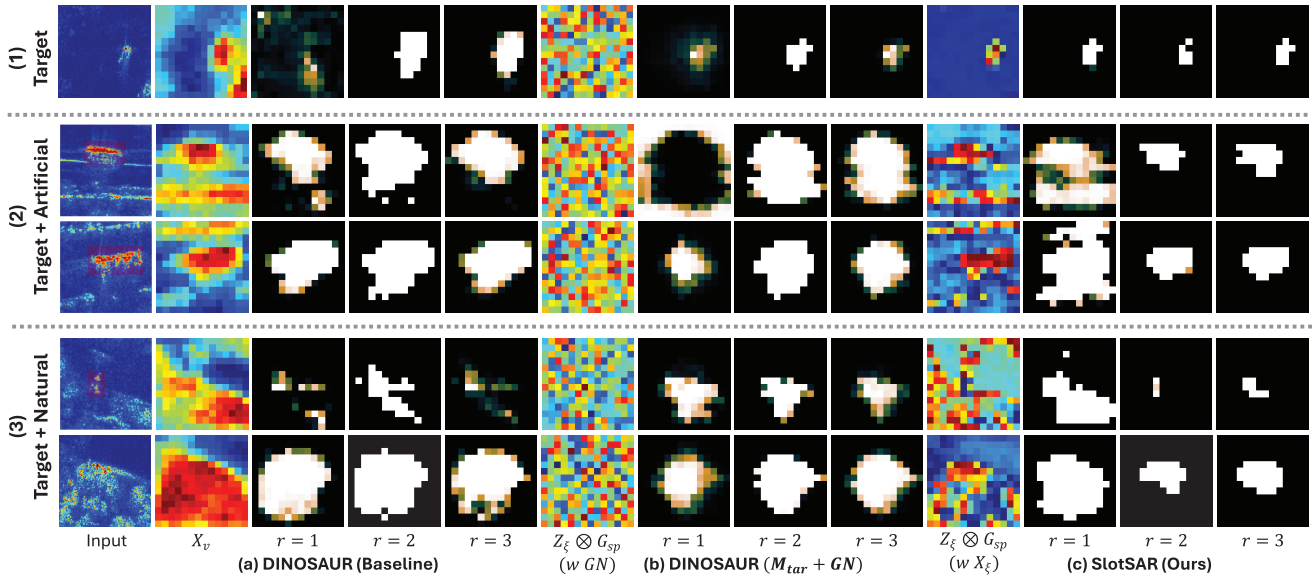


Fig. 13. Qualitative comparison of feature maps and slot attention maps across models and iterations. (a) In DINOSAUR, attention fails to converge even as r increases. (b) GN produces uninformative fusion maps with little change across iterations. (c) SlotSAR progressively refines attention toward regions where semantic and scattering cues overlap, effectively separating targets from clutter.

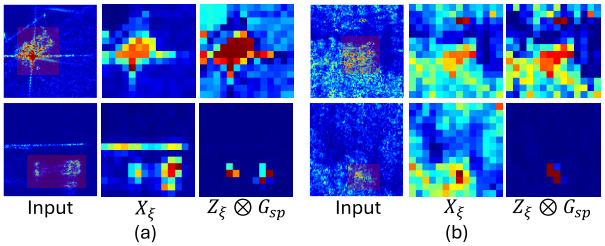


Fig. 14. Qualitative comparison of scattering features X_ξ and refined features $Z_\xi \otimes G_{sp}$ under (a) artificial clutter and (b) natural clutter conditions.

TABLE VII

QUANTITATIVE ANALYSIS OF TARGET AND CLUTTER ENERGY FOR X_ξ AND $Z_\xi \otimes G_{sp}$ ACROSS EOC-S

EOC-S	Feature	$E_{\text{tar}} \uparrow$	$E_{\text{clu}} \downarrow$	$E_{\text{tar}}/E_{\text{clu}} \uparrow$
Test	X_ξ	0.65	0.35	1.86
	$(Z_\xi \otimes G_{\text{sp}})$	**0.69**	**0.31**	**2.26**
City	X_ξ	0.73	0.27	2.69
	$(Z_\xi \otimes G_{\text{sp}})$	**0.76**	**0.24**	**3.19**
Factory	X_ξ	0.69	0.31	2.18
	$(Z_\xi \otimes G_{\text{sp}})$	**0.70**	**0.30**	**2.37**
Woodland	X_ξ	0.82	0.18	4.52
	$(Z_\xi \otimes G_{\text{sp}})$	**0.84**	**0.16**	**5.34**

depending on the clutter environment. After spatial weighting, $Z_\xi \otimes G_{\text{sp}}$ exhibits higher target energy and lower clutter energy across all environments, resulting in consistently higher energy ratios than X_ξ. These results indicate that GFRM spatially refines the scattering responses captured by WSN, increasing their concentration within the candidate regions for subsequent object grouping.

Slot Convergence: Having shown that GFRM incorporates local scattering responses and their spatial distributions into the value representation, we further examine how consistently target-related patterns are represented across different backgrounds and whether these patterns are progressively utilized during iterative slot updates. To this end, we construct recurrence maps that identify feature patterns repeatedly observed across different SAR images. Patch features from the entire EOC test set are jointly clustered into $K = 128$ clusters using K-means, and each cluster is assigned a recurrence score based on the proportion of images in which it appears. To reduce the influence of common background patterns that occupy broad regions, we penalize clusters according to their average spatial coverage within each image. Each patch is then assigned a value based on the recurrence score of its cluster and its similarity to the corresponding cluster center, and these values are rearranged into a recurrence map. Accordingly, higher values in the recurrence map indicate feature patterns that are more consistently observed across different images, whereas lower values indicate less recurrent patterns.

As shown in Fig. 15(a), the recurrence map constructed from the scattering features X_ξ achieves an mBO of 21.18% with the ground-truth target regions, which increases to 28.56% when visual and scattering information are jointly considered through $X_\xi \otimes G_{sp}$. The recurrence map visualizations in Fig. 16 also show that recurring feature responses are primarily concentrated around the target regions. These observations support our motivation that target-related feature patterns remain relatively consistent across different images, whereas clutter responses vary with the surrounding background. Moreover, the increased mBO for $X_\xi \otimes G_{sp}$ indicates that jointly considering visual and scattering features provides more target-related recurring patterns.

We next examine whether these patterns are progressively utilized during slot attention. As shown in Fig. 15(b), the mBO between the recurrence and slot attention maps increases from 14.94% to 17.22% and 18.40% for X_ξ, and from 21.37%

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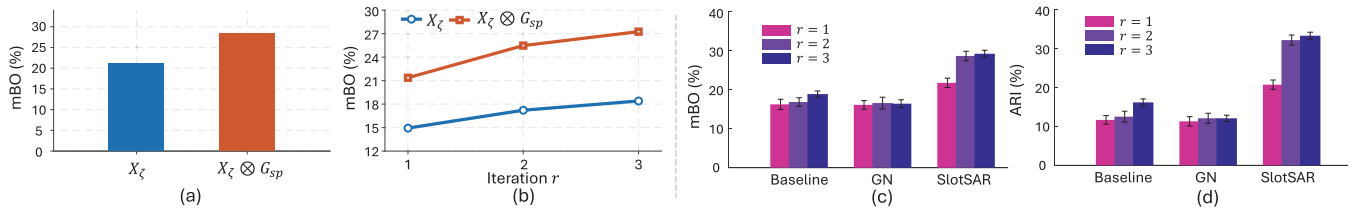


Fig. 15. Quantitative analysis of feature recurrence and iterative slot convergence. (a) mBO between recurrence maps and ground-truth target regions. (b) mBO between recurrence and slot attention maps across slot iterations. (c) mBO and (d) ARI performance across iterative slot updates.

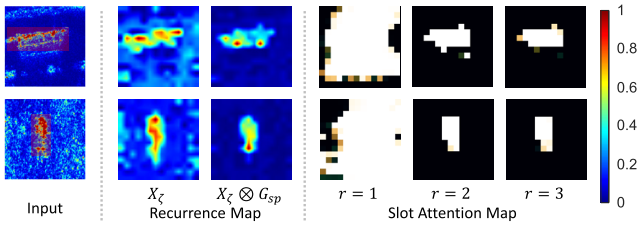


Fig. 16. Qualitative visualization of recurrence maps and slot attention maps across iterative slot updates.

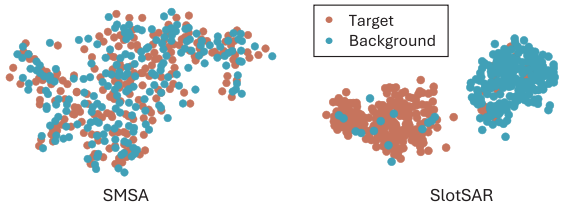


Fig. 17. t-SNE visualizations of target and clutter slots for SMSA [61] and SlotSAR.

to 25.48% and 27.26% for $X_\xi \otimes G_{sp}$ over three iterations. This increasing correspondence indicates that slot attention progressively aligns with recurring visual–scattering patterns during iterative updates. A similar tendency is qualitatively observed in Fig. 13: the target and clutter regions remain substantially intermixed for DINOSAUR, whereas SlotSAR forms more target-consistent attention maps. Furthermore, as shown in Fig. 15(b) and (c), ARI and mBO improve with increasing iterations for SlotSAR, whereas replacing X_ξ with Gaussian noise (GN) results in only limited improvements. Together, these results suggest that the target-related visual–scattering information incorporated by GFRM is grouped into object-centric representations through iterative slot updates.

5) Slot Representation Analysis: To verify how effectively target and clutter are separated across slots, we visualize the features of each slot using t-SNE [66] and compare them with recent slot attention-based methods [61]. As shown in Fig. 17, SMSA [61] exhibits significant overlap among slot distributions, consistent with its poor quantitative performance in Table III. This indicates that applying vector quantization in SMSA causes multiple slots to be mapped to shared codebook entries, weakening slot competition and forcing them to attend to identical spatial regions. In contrast, SlotSAR demonstrates clear slot-wise disentanglement in the latent space and superior OCL performance. This suggests that the incorporation of WSN and the feature refinement performed by GFRM strengthen competition among slots, leading to improved separation between target and clutter despite their high-level feature similarity in SAR imagery.

D. Evaluation of ATR

Following the evaluation of SlotSAR, we proceed to assess the ATR performance of the SlotATR by conducting classification experiments with SlotCLS. This allows us to evaluate the effectiveness of the learned object-centric representations for downstream ATR classification. All results are reported as the average of five independent runs to ensure statistical reliability, with the best and second-best performances highlighted in bold and underlined, respectively.

1) Performance Comparison With ATR Methods: To evaluate the effectiveness of the SlotATR, we compare it with representative SAR ATR methods employing different strategies for clutter reduction, clutter-robust learning, and ASC-based target representation. First, we consider the preprocessing-based clutter reduction method ESENet [62] and FLCR methods, including CRM [16] and HDANet [17]. ESENet extracts target regions based on image intensity and generally assumes that the target is located near the center of the SAR image during preprocessing [20], [21]. However, since targets in the ATRNet-STAR dataset are not always centered [9], preprocessing-based target extraction is performed over the entire image without using any location prior. CRM learns FLCR using pseudo ground-truth generated from preprocessing-based target extraction [16], whereas HDANet employs Grad-CAM-based saliency maps as supervision [17]. We also compare with background robust learning (BRL [26]), which improves robustness against background variations using intensity-based target priors, and the SAR foundation model SARATRX [12]. In addition, the bounding-box annotations provided in the ATRNet-STAR dataset allow the generation of ground-truth target masks [9]. To analyze the effect of explicit target localization, we further evaluate HDANet and BRL using ground-truth target masks, denoted as HDANet-GT and BRL-GT, respectively. For a fair comparison, all methods are implemented using the same ConvNext backbone and trained and evaluated under identical experimental settings.

As shown in Table VIII, SlotATR achieves the highest average recognition performance across all EOC-S settings on the ATRNet-STAR ground-range dataset. In contrast, ESENet and CRM exhibit inferior performance compared with the other methods. Since both methods rely heavily

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TABLE VIII

COMPARISON OF CLASSIFICATION ACCURACY (%) ON THE ATRNET-STAR GROUND-RANGE DATASET UNDER THE EOC-S SETTING

Model	Test	Factory	City	Woodland	Avg.
ConvNext	27.12	28.23	25.67	24.73	26.44
ESENet	14.14	16.58	11.89	12.34	13.74
SARATRX	19.50	20.59	20.27	19.44	19.95
ConvNext+CRM	21.57	22.71	20.80	20.56	21.41
HDANet	31.62	32.43	31.69	28.19	30.98
HDANet-GT	32.68	33.19	32.50	30.62	32.25
BRL	31.82	31.98	31.53	33.61	32.24
BRL-GT	36.20	37.01	**36.22**	34.40	35.96
SlotATR	**37.44**	**41.85**	35.88	**35.56**	**37.68**

TABLE IX

COMPARISON OF CLASSIFICATION ACCURACY (%) ON THE ATRNET-STAR SLANT-RANGE DATASET UNDER THE EOC-S SETTING

Model	Test	Factory	City	Woodland	Avg.
ConvNext	20.55	22.02	21.06	12.72	19.09
SANet	25.49	28.21	24.11	16.06	23.47
ASC-SepNet	27.92	30.24	25.31	28.64	28.03
VSFA*	16.98	19.14	16.45	15.25	16.96
LDSF*	30.52	32.00	31.63	27.92	30.52
SARGTR*	31.16	33.38	**32.69**	27.40	31.16
SlotATR	**35.15**	**39.65**	32.26	**28.69**	**33.94**

on preprocessing-based target extraction, they fail to reliably isolate target regions under severe clutter conditions. This limitation becomes more pronounced in the EOC-S setting, where the clutter distributions between the training and testing environments differ substantially, resulting in a significant degradation in recognition performance. In contrast, HDANet and BRL achieve higher recognition performance than preprocessing-based methods by exploiting feature-level supervision and background robustness, respectively. Furthermore, HDANet-GT and BRL-GT, which utilize ground-truth target masks, further improve upon their corresponding baseline models. This indicates that explicit target localization facilitates the learning of target-centric representations while also demonstrating that these methods benefit from target-mask supervision. Nevertheless, SlotATR achieves competitive performance with HDANet-GT and BRL-GT without requiring explicit target-mask annotations. These results demonstrate that OCL can effectively separate target and clutter regions while learning discriminative target representations. Finally, SlotATR surpasses the large-scale SAR foundation model SARATRX. Although SlotATR is built upon features extracted from SARATRX, incorporating OCL leads to substantial performance gains under the EOC-S setting. This suggests that, when the clutter distributions between the training and testing environments differ significantly, refining pretrained representations into target-centric representations through OCL is more effective than directly utilizing the pretrained features.

To further investigate the effectiveness of the SlotATR, we compare it with ASC [23]-based SAR ATR methods. ASC-based methods represent SAR targets by exploiting their physical scattering characteristics and require complex-valued data containing both amplitude and phase information [22], [23]. For a fair comparison, all methods are trained and evaluated on the ATRNet-STAR EOC-S slant-range dataset using the same ConvNext backbone, as summarized in Table IX. Results marked with an asterisk (*) are reported from the original papers. As shown in Table IX, most ASC-based methods outperform the baseline ConvNext, demonstrating that ASC information provides effective target representations under severe clutter conditions. SlotATR achieves the highest average recognition performance among all ASC-based methods. While ASC-based methods explicitly extract scattering centers to exploit the physical characteristics of SAR targets, SlotATR directly learns target-centric representations by separating target and clutter regions at the feature level through OCL. These results suggest that OCL enables effective target-centric representation learning for SAR ATR in severe clutter conditions.

TABLE X

COMPARISON OF CLASSIFICATION ACCURACY (%) ON THE MSTAR DATASET UNDER THE SOC SETTING

Methods	ESENet	SARATRX	HDANet	BRL	ASC-SepNet	SlotATR
Acc. [%]	97.32	**99.64**	**99.64**	99.58	**99.64**	99.58

Finally, to evaluate the generalization performance of SlotATR on a different dataset, we compare its ATR performance on the MSTAR dataset. As shown in Table X, SlotATR achieves recognition performance comparable to existing methods, showing that SlotATR generalizes well across different SAR datasets. Notably, its advantage becomes more evident under the EOC-S setting of the ATRNet-STAR dataset, where severe clutter variations make robust target-centric representation learning more important.

2) Ablation Study: To further validate SlotATR, we conduct comprehensive ablation studies by comparing different OCL frameworks, evaluating the generalization of SlotATR across multiple ATR backbones, analyzing the contribution of key components, and comparing with end-to-end training. The OCL framework comparison and backbone generalization experiments are conducted using VGG16 [67], ResNet [68], and ConvNext [69] under the EOC-S setting of the ATRNet-STAR dataset.

Different OCL Modules in SlotATR: To further investigate the impact of OCL on classification performance, we conduct experiments by replacing the OCL module within SlotCLS with different methods. Table XI summarizes the ATR results obtained using various OCL approaches across multiple backbone models. First, incorporating OCL-based feature refinement consistently improves performance over standard ATR training, demonstrating that effective clutter reduction is particularly beneficial in the EOC-S setting, where there exists a significant mismatch between training and testing clutter distributions. Second, among the evaluated OCL methods, SlotSAR achieves the best average accuracy across all backbones and generally shows superior performance in both artificial and natural clutter conditions. These results indicate that higher-quality object-centric representations lead to more effective clutter reduction and, consequently, improved downstream ATR performance.

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TABLE XI

PERFORMANCE COMPARISON OF DIFFERENT OCL METHODS WITHIN SLOTCLS ACROSS MULTIPLE BACKBONES UNDER THE EOC-S SETTING

Model	Method	Test	Factory	City	Woodland	Avg.
VGG16	Baseline	21.60	20.50	22.70	6.80	17.90
	DINOSAUR	25.89	27.44	25.10	22.39	25.21
	ISA	24.78	29.82	21.83	23.89	25.08
	SlotSAR	**28.72**	**33.54**	**25.35**	**25.68**	**28.32**
ResNet18	Baseline	16.10	13.70	18.10	4.20	13.03
	DINOSAUR	24.90	27.21	23.40	22.44	24.49
	ISA	25.26	29.65	22.50	19.45	24.22
	SlotSAR	**29.47**	**34.54**	**26.78**	**27.23**	**29.51**
ResNet34	Baseline	18.00	16.80	18.90	5.40	14.78
	DINOSAUR	25.82	29.53	23.56	19.45	24.59
	ISA	24.87	29.05	21.87	**23.60**	24.85
	SlotSAR	**28.95**	**34.11**	**26.73**	23.27	**28.27**
ConvNext	Baseline	27.12	28.23	25.67	24.73	26.44
	DINOSAUR	30.14	33.95	28.01	21.50	28.40
	ISA	29.57	33.35	26.78	28.49	29.55
	SlotSAR	**37.44**	**41.85**	**35.88**	**35.56**	**37.68**

TABLE XII

COMPARISON OF SLOTCLS WITH CBAM ACROSS MULTIPLE ATR BACKBONES UNDER THE EOC-S SETTING

Model	Method	Test	Factory	City	Woodland	Avg.
VGG16	Baseline	21.60	20.50	22.70	6.80	17.90
	CBAM	23.73	20.93	**25.84**	23.79	23.57
	SlotCLS	**28.72**	**33.54**	25.35	**25.68**	**28.32**
ResNet18	Baseline	16.10	13.70	18.10	4.20	13.03
	CBAM	18.00	16.98	19.05	15.09	17.28
	SlotCLS	**29.47**	**34.54**	**26.78**	**27.23**	**29.51**
ResNet34	Baseline	18.00	16.80	18.90	5.40	14.78
	CBAM	20.79	19.29	22.09	19.20	20.34
	SlotCLS	**28.95**	**34.11**	**26.73**	**23.27**	**28.27**
ConvNext	Baseline	27.12	28.23	25.67	24.73	26.44
	CBAM	26.20	26.79	26.19	21.65	25.21
	SlotCLS	**37.44**	**41.85**	**35.88**	**35.56**	**37.68**

Generalization Across ATR Backbones: To validate the generalizability of the proposed method, we compare SlotATR with the baseline and a widely used attention-based module, CBAM [70], which has also been widely adopted in SAR ATR tasks [27], across multiple ATR backbones under the EOC-S setting. The results are summarized in Table XII. Across all backbones and evaluation conditions, SlotATR consistently outperforms both the baseline and CBAM. In particular, significant performance gains are observed under severe clutter conditions, indicating that explicitly reducing clutter interference is critical in the EOC-S setting. Compared to CBAM, which enhances features through generic attention mechanisms, SlotATR achieves superior performance by utilizing object-centric representations that explicitly separate target and clutter. This enables more reliable focus on target regions, leading to improved classification accuracy across diverse environments. These results suggest that, in clutter-dominated scenarios, effective mitigation of clutter interference and accurate attention to target regions are essential for improved ATR performance, with SlotATR providing a practical approach across different backbone architectures.

Ablation Study on SlotATR Components: To analyze the contribution of each component in SlotATR, we conduct an ablation study under the EOC-S setting using a ResNet18 backbone. The results are summarized in Table XIII, where we evaluate the impact of each SlotCLS component (target selector, magnitude map, and channel map) based on the slot attention map generated by SlotSAR.

TABLE XIII

ABLATION STUDY ON SLOTATR COMPONENTS UNDER RESNET18 BACKBONE

Index	SlotSAR	SlotCLS			EOC-S
	Slot Attn Map	Target Selector	Spatial Attn Map	Channel Attn Map	
Baseline					16.10
(1)	✓				18.38
(2)	✓	✓			27.88
(3)	✓		✓		18.34
(4)			✓		16.78
(5)	✓	✓	✓		28.96
(6)	✓	✓		✓	28.95
(7)	✓		✓	✓	19.95
(8)	✓	✓	✓	✓	**29.47**

Starting from the baseline, incorporating only the slot attention map (1) yields limited performance improvement, indicating that attention alone is insufficient for effective target–clutter discrimination. Similarly, combining the attention map with either the magnitude map (3) or the channel map (7) results in only marginal gains. This suggests that additional feature refinement is ineffective when target regions are not explicitly identified.

In contrast, introducing the target selector (2) leads to a substantial performance improvement, demonstrating that explicitly selecting target-relevant regions is critical for effective representation learning. Further improvements are achieved when refinement mechanisms are applied to the selected regions. Specifically, incorporating the magnitude map (5) or the channel map (6) provides additional gains, indicating that emphasizing salient spatial responses or reweighting channel-wise features enhances discriminability. Finally, combining all components (8) achieves the best performance, confirming that modeling target selection and feature refinement is essential for maximizing performance under severe clutter conditions.

Impact of Training Strategy: To investigate the effect of joint and separate optimization, we compare two training strategies for SlotATR: 1) joint training, in which SlotSAR and SlotCLS are optimized simultaneously, and 2) two-stage training, in which SlotSAR is first trained and SlotCLS is subsequently optimized based on the learned SlotSAR. Both strategies are evaluated under the EOC-S setting of the ATRNet-STAR dataset. The OCL performance is evaluated using the generated slot attention maps, while the ATR performance is evaluated using a ConvNext backbone. In joint training, SlotSAR is optimized with the reconstruction loss, while the classification loss of SlotCLS, which takes the slot attention maps generated by SlotSAR as input, is jointly optimized. In contrast, two-stage training first trains SlotSAR using only the reconstruction loss, after which SlotCLS is trained separately using the slot attention maps generated by the learned SlotSAR. The comparison results are summarized in Table XIV.

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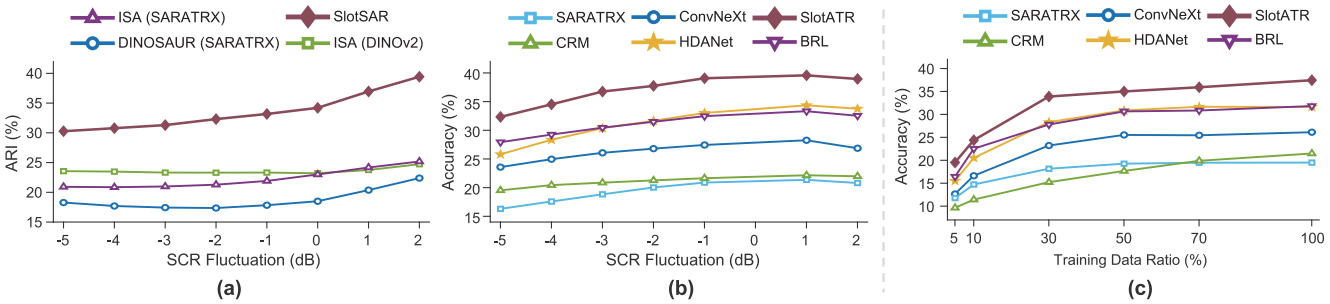


Fig. 18. (a) ARI of different OCL methods under SCR fluctuations. (b) Classification accuracy of different ATR methods under SCR fluctuations. (c) Classification accuracy of models trained with varying proportions of the training data.

TABLE XIV
COMPARISON OF JOINT AND TWO-STAGE TRAINING STRATEGIES

Setting	OCL		ATR
	ARI [%] ↑	mBO [%] ↑	Acc. [%]
Joint Training	27.38	22.17	30.15
Two-stage Training	**33.36**	**29.19**	**37.44**

As shown in Table XIV, two-stage training consistently outperforms joint training in both ARI and mBO, indicating that independently optimizing SlotSAR improves its ability to learn object-centric representations. One possible explanation is that the classification objective encourages slot representations to capture class-discriminative information, which may not fully align with the objective of separating target and clutter regions. Consequently, classification gradients may interfere with object-centric representation learning, resulting in less effective target–clutter separation. In contrast, two-stage training allows SlotSAR to first learn object-centric representations solely through the reconstruction objective and subsequently uses the learned representations for ATR classification. This advantage is also reflected in the downstream ATR performance, where two-stage training achieves higher classification accuracy than joint training. These results suggest that separately optimizing SlotSAR before classifier training is more effective under severe clutter conditions.

3) Robustness Under SCR Variations: To evaluate the robustness of SlotATR against signal-to-clutter ratio (SCR) fluctuations, experiments were conducted based on the EOC-S setting of the ATRNet-STAR dataset. All models were trained using only the original EOC-S training set (0 dB), while evaluation was performed on test sets with different SCR levels. The SCR-fluctuated test images were generated by preserving the target region using the ground-truth target mask and adjusting only the clutter intensity, with SCR values ranging from −5 to +2 dB. This experimental setting enables the evaluation of SlotATR under unseen SCR conditions.

The ARI results under different SCR fluctuations are presented in Fig. 18(a). Most methods exhibit improved ARI as the SCR increases, whereas SlotSAR consistently achieves the highest ARI across all SCR fluctuations. In particular, SlotSAR maintains superior performance under low-SCR conditions, demonstrating its ability to learn effective object-centric representations even in severe clutter environments.

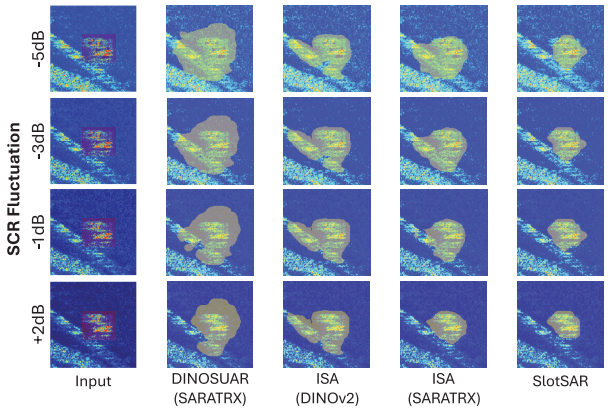


Fig. 19. Qualitative comparison of target attention maps generated by different OCL methods under varying SCR fluctuation conditions. Yellow regions represent the target areas predicted by each model, and red regions indicate the ground-truth target regions.

The target attention maps in Fig. 19 further illustrate the effectiveness of SlotSAR under SCR fluctuations. Existing OCL methods tend to activate clutter regions together with the target or produce ambiguous target boundaries as the SCR decreases. In contrast, SlotSAR consistently produces more discriminative target attention maps while reducing clutter activation across different SCR levels. These qualitative observations are consistent with the ARI results and further demonstrate the effectiveness of the proposed object-centric representations under varying SCR conditions.

The ATR performance under different SCR fluctuations is presented in Fig. 18(b). SlotATR achieves the highest classification accuracy across all SCR levels, including low-SCR conditions. These results indicate that the target-centric attention maps generated by SlotSAR effectively reduce the influence of clutter under challenging SCR conditions, thereby contributing to improved ATR performance.

4) Few-Shot Evaluation: To evaluate the performance of the SlotATR under limited training data, experiments were conducted by varying the proportion of the training set on the EOC-S setting of the ATRNet-STAR dataset. All experiments were performed using ConvNext as the backbone, while the test set was kept unchanged. The training set was randomly sampled at 5%, 10%, 30%, 50%, 70%, and 100% of the full

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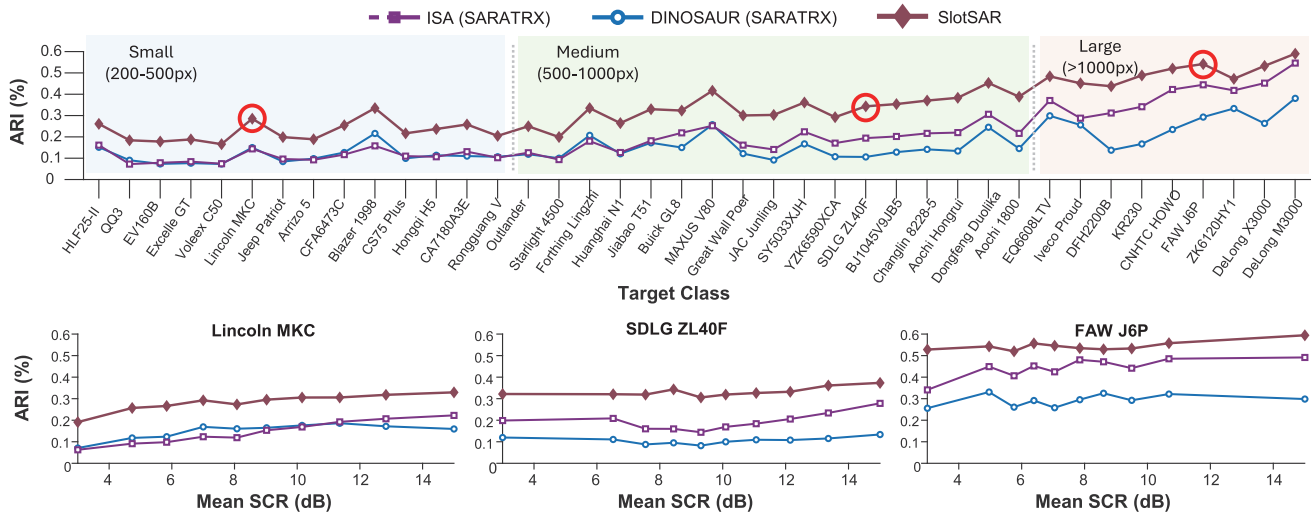


Fig. 20. Class-wise OCL performance across different target sizes. Target classes are sorted in ascending order according to the average ground-truth bounding-box area. The upper plot presents the ARI of each target class, while the lower plots show the ARI under varying mean SCR conditions for representative classes from each target-size group: Lincoln MKC (Small), SDLG ZL40F (Medium), and FAW J6P (Large).

TABLE XV

COMPARISON OF ATR PERFORMANCE ACROSS DIFFERENT TARGET SIZES. TARGET CLASSES ARE CATEGORIZED INTO SMALL (200–500 PX), MEDIUM (500–1000 PX), AND LARGE (>1000 PX) ACCORDING TO THE AVERAGE GROUND-TRUTH BOUNDING-BOX AREA

Model	Overall			Small			Medium			Large		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
ConvNext	27.22	41.16	27.22	15.67	17.19	14.99	17.88	46.27	17.68	48.52	59.62	49.71
SARATRX	19.50	25.34	19.43	14.50	9.38	13.85	11.56	24.91	11.55	35.86	43.27	37.36
ConvNext+CRM	21.57	31.61	21.62	15.20	7.05	14.55	9.03	30.71	8.99	41.31	57.13	42.30
BRL	31.82	**44.26**	31.72	14.51	18.53	13.88	24.25	**51.13**	23.94	56.79	**62.58**	57.93
HDANet	31.62	43.01	31.53	13.68	16.73	13.24	22.33	49.24	21.93	59.65	**62.58**	60.52
SlotATR	**37.44**	42.30	**37.33**	**17.95**	**29.31**	**17.55**	**32.42**	49.95	**32.30**	**62.62**	54.61	**63.65**

training data. To ensure a fair comparison, the sampled training subset for each ratio was fixed and shared across all methods.

The ATR performance under different training data ratios is presented in Fig. 18(c). As the amount of training data increased, all methods exhibited improved classification performance. Compared with the ConvNext baseline, HDANet and BRL consistently achieved higher classification accuracy across all training data ratios, showing the effectiveness of clutter mitigation under limited training data. In contrast, CRM showed relatively lower performance, which is attributed to its reliance on target masks generated through a preprocessing stage. Consequently, the quality of the generated masks has a greater impact when only limited training data are available. The SlotATR achieved the highest classification accuracy across all training data ratios. These results demonstrate that the OCL-based ATR framework provides improved classification performance compared with existing methods, even under limited training data conditions.

5) Analysis Across Target Sizes: To analyze the effectiveness of the SlotATR framework across different target sizes, both OCL and ATR performance are evaluated.

1) *OCL Performance:* Fig. 20 presents the OCL performance of various methods across different target sizes on the EOC-S test set. The target size of each class was computed using the ground-truth bounding boxes, and the classes were sorted in ascending order according to their average target size. In addition, representative classes from each target-size group were selected to compare the performance under varying mean SCR conditions. As shown in Fig. 20, most methods exhibited degraded OCL performance as the target size decreased, suggesting that target-clutter separation becomes more challenging for smaller targets. Nevertheless, SlotSAR achieved higher ARI than the existing OCL methods for most target classes. A similar trend was observed across varying mean SCR conditions. Specifically, SlotSAR consistently achieved the highest ARI across all SCR levels for Lincoln MKC (Small), SDLG ZL40F (Medium), and FAW J6P (Large). These results show that SlotSAR provides better OCL performance than the existing methods for SAR targets of different sizes.
2) *ATR Performance:* Table XV presents the ATR performance across different target sizes. As shown in Fig. 20, all target classes were categorized into three groups based on the average area of their ground-truth bounding boxes: small (200–500 px), medium (500–1000 px), and large (above 1000 px). As shown in Table XV, most methods, including clutter mitigation approaches, exhibited degraded ATR performance

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as the target size decreased. This suggests that small targets remain challenging to recognize. Despite this trend, SlotATR achieved the highest accuracy across all three target-size groups, including small, medium, and large. These results demonstrate that SlotATR provides improved classification performance over existing clutter mitigation methods for SAR targets of different sizes.

V. Discussion

A. Model Size

DINOSAUR methods consist of approximately 50.46 M parameters in the visual encoder and 1.71 M parameters in the slot attention module. Unlike the baseline methods, SlotSAR additionally incorporates a scattering encoder. However, this module contains only 64 parameters, resulting in a negligible impact on the overall model size. Furthermore, the additional parameters introduced by GFRM amount to only about 0.5 M. Meanwhile, the modules included in SlotCLS (comprising embedding, MLP, and spatial map extraction) require approximately 30 K parameters per module. These results demonstrate that the proposed design introduces minimal parameter overhead while effectively improving target–clutter disentanglement in SAR imagery, as well as enhancing class-level discrimination within the extracted target regions.

B. Limitations and Future Directions

Despite the promising performance achieved by SlotATR, it is important to acknowledge that several limitations still remain. We believe that these limitations primarily lie in the following two aspects.

1) Low-SCR and Small-Target Scenarios: Although SlotATR achieves superior performance compared with existing SAR-based OCL and clutter mitigation methods, it still exhibits limitations under low-SCR conditions and for small targets. As shown in Fig. 20 and Table XV, the proposed framework consistently outperforms existing methods across all target-size groups. However, the absolute OCL and ATR performance for small targets remains noticeably lower than that for medium and large targets. This limitation can be attributed to the fact that small targets are represented by only a limited number of feature tokens in the feature map.

Under low-SCR conditions, a single feature token is more likely to contain both target and surrounding clutter information, requiring slot attention to disentangle mixed representations. This is consistent with the well-known structural limitation of slot attention-based models, where small objects are difficult to separate reliably under limited feature resolution [71]. SlotSAR alleviates this limitation by refining the visual features through $Z_\xi \otimes G_{sp}$. However, the same limitation may reemerge when the scattering features are also represented by low-resolution feature tokens, similar to the visual features. As illustrated in Fig. 21, this limitation can result in partial overlap between the predicted target region and surrounding clutter. This degrades the quality of object-centric representations and consequently affects the ATR performance.

In future work, we plan to address this limitation by developing high-resolution object-centric representation learning for preserving fine-grained scattering structures of small targets. In particular, we will investigate graph-based representations that model scattering points as graph nodes and explicitly learn their structural relationships, enabling more effective preservation of small-target structures even under severe clutter conditions.

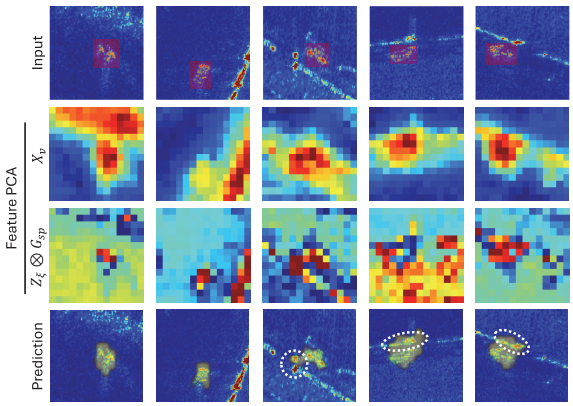


Fig. 21. Visualizations of input images, feature maps, and prediction outputs for small targets under cluttered conditions. The red boxes denote the ground-truth target regions.

2) Multitarget Scenarios: The proposed SlotATR is designed to mitigate clutter interference during target recognition in postdetection single-target SAR imagery. Accordingly, it learns object-centric representations to disentangle a single target from the background clutter, while multitarget scenarios are beyond the scope of this work. One promising direction for improvement would be to develop a unified object-centric SAR framework for joint target detection and recognition in multitarget scenes. Specifically, scattering-aware representations could be integrated with object-centric features to assign multiple targets to individual slots, while set prediction could be incorporated for end-to-end multiobject recognition.

VI. Conclusion

In this study, we proposed SlotATR, an OCL-based framework for SAR target recognition that aims to mitigate clutter interference and learn target-centric representations without requiring target-mask annotations. Unlike conventional approaches that rely on preprocessing or supervised clutter reduction, SlotATR addresses clutter interference through object-centric representation learning in the feature space. SlotATR consists of two stages: SlotSAR and SlotCLS. SlotSAR complements high-level visual features with local scattering responses and their spatial distributions to facilitate object grouping. Through iterative slot updates, recurring visual and scattering patterns are grouped into object-centric representations of the target and clutter. SlotCLS subsequently incorporates the learned target-centric representation into ATR backbones for discriminative classification. Experimental results demonstrate that SlotATR consistently improves ATR performance, particularly under varying clutter conditions. Moreover, SlotATR achieves competitive performance without explicit target-mask supervision, reducing reliance on costly pixel-level annotations.

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Although the proposed framework demonstrates promising performance, there is still room for improvement. Future work will focus on improving its robustness and extending the framework to multitarget SAR scenarios.

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